

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

Mahimn Patel
Department of Economics, University of Toronto
`mahimn.patel.k@gmail.com`

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, derived from the open-interest grid via a Gârleanu–Pedersen–Potesman (Gârleanu et al., 2009) demand-pressure mapping. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable z_t —and reduces to standard time-dependent Heston (Heston, 1993) at zero coupling. Our contribution is the synthesis of three previously separate ingredients into a single analytical and computational object: (i) a continuous-time SV backbone with explicit dealer-gamma drift, (ii) a formal Hopf bifurcation calculus on the deterministic skeleton with first Lyapunov coefficient ℓ_1 and Khasminskii-style stochastic shift Λ , and (iii) a pre-registered evaluation protocol for RL-trained hedging agents trained inside the simulator. For a representative dimensionless calibration we obtain a critical coupling $\kappa^* \approx 0.8964$, angular frequency $\omega^* \approx 0.5724$ rad/yr, and first Lyapunov coefficient $\ell_1 \approx -0.025 < 0$, so the bifurcation is supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$. The 2D reduction (without z_t) has an upper-triangular Jacobian and admits only a saddle-node onset, recovering the discrete-time recursion of Dai (2025); the 3D extension is what unlocks the Hopf. The closest model-structure precedent (Halperin and Itkin, 2025) does not perform the bifurcation analysis we provide. For the natural parametric specification with log-normal open-interest in moneyness, we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space.

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. (2013), who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros (2009) and Bouchaud (2011) are the conceptual antecedents; Filimonov and Sornette (2012) formalised the reflexivity-quantification programme that Hardiman et al. (2013) executed. The question we take up here is what continuous-time mechanism produces this critical reflexivity, and—given a candidate mechanism—how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma feedback derived from the option open-interest grid via the Gârleanu et al. (2009) demand-pressure mapping. The variance backbone is Heston (Heston, 1993); the memory equation $\dot{z} = -\alpha z + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion (Routh–Hurwitz with one zero), with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 14–15).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (sliced-Wasserstein-2 over 21-day surface windows, κ -sensitivity slope, Hopf-frequency spectral test) at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. (2018) adapted for computational finance.

Closest precedents. Halperin and Itkin (2025) (Marketron) share the model-structural axis—continuous-time SDE, memory channel, options-aware—but do not perform the bifurcation analysis. Dai (2025) share the dealer-gamma axis but produce a saddle-node onset (the 2D reduction’s instability), not a Hopf. He et al. (2025) (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents (He et al., 2009; Chiarella et al., 2006) use delay-differential heterogeneous-belief models without an options-grid feedback channel; the discrete-time Brock et al. (2009) “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker (Hopf-for-maps) bifurcations. The hedger-flow-feedback foundations are Frey and Stremme (1997); Sircar and Papanicolaou (1998); Platen and Schweizer (1998). Sections 3, 5 and 6 make these contrasts precise.

Roadmap. Section 2 states the model. Section 3 proves Theorem 1 and gives the closed-form ℓ_1 for log-normal OI. Section 4 reports the numerical phase diagram. Section 5 describes the pre-registered evaluation framework. Section 6 reports the mechanism decomposition against Marketron. Section 7 closes with the empirical-calibration roadmap. Code at <https://github.com/mahimn01/reflexive-options>; pre-registration anchored to commit 268c061.

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and z_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

$$d\langle W^S, W^v \rangle_t = \rho dt, \quad (4)$$

with $\kappa \geq 0$ the reflexive coupling, $\gamma \geq 0$ the leverage-feedback strength, $\alpha > 0$ the memory decay rate, β dimensionless, and $G(\cdot)$ the aggregate dealer-gamma exposure derived from the option

open-interest grid:

$$G(S, t) = \sum_{K, T} q_{K, T} \Gamma_{K, T}(S, t) \text{sign}(K, T), \quad (5)$$

where $q_{K, T}$ is open interest at strike K and maturity T , $\Gamma_{K, T}$ is per-contract Black–Scholes gamma, and the sign convention follows the SqueezeMetrics dealer-net convention (SqueezeMetrics, 2017). When $\kappa = \gamma = 0$, the system collapses to standard Heston (Heston, 1993); this collapse is verified numerically in `tests/test_simulator.py`.

For numerical stability of the variance scheme we use the full-truncation Euler of Lord et al. (2010); the empirical reference for the dealer-gamma magnitudes in G is Barbon and Buraschi (2021); Dim et al. (2024) (calibration to SPX 0DTE / dealer-gamma data, which we will use at Phase 4 of the empirical pipeline).

Why three states. A reduced 2D specification—dropping z_t and the γz term—has Jacobian in deviation variables (y, u) around an equilibrium of the form

$$J_{2D}(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix},$$

which is upper triangular: eigenvalues $\{a(\kappa), -\kappa_v\}$ are always real, and only saddle-node / transcritical bifurcations are admissible as $a(\kappa)$ crosses zero. This recovers the static onset of Dai (2025). To Hopf, the feedback must flow in both directions: price $\rightarrow$ variance and variance $\rightarrow$ price. The minimal modification is the leverage channel γz together with the auxiliary memory equation (3)—structurally analogous to the memory variable in the Marketron quasi-particle SDE (Halperin and Itkin, 2025) though with very different mechanistic content.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria and Jacobian

Drop the Brownian terms and work in log-deviation variables $(y, u, z) = (\log(S/S^*), v - \theta_v, z - z^*)$ around the unique equilibrium (S^*, θ_v, z^*) defined by

$$\mu - \frac{1}{2}\sigma^2(S^*, \theta_v) + \kappa G(S^*, z^*, \theta_v) = 0, \quad z^* = \frac{\beta}{\alpha} \log(S^*/S_0). \quad (6)$$

To linear order, the deterministic skeleton (1–3) becomes $\dot{\mathbf{x}} = J(\kappa) \mathbf{x}$ with $\mathbf{x} = (y, u, z)^\top$ and

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_z \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (7)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$, and G_x, G_v, G_z denote partial derivatives of G at the equilibrium.

3.2 Routh–Hurwitz / Liu’s criterion

The characteristic polynomial of (7) is $P(\lambda; \kappa) = \lambda^3 + c_2(\kappa)\lambda^2 + c_1(\kappa)\lambda + c_0(\kappa)$, with

$$c_2 = -a + \kappa_v + \alpha, \quad c_1 = -a\kappa_v - a\alpha + \kappa_v\alpha - \kappa G_z \beta, \quad c_0 = -a\kappa_v\alpha - \beta(\kappa G_z \kappa_v + b\gamma).$$

A 3D system Hopfs iff (Liu’s criterion, equivalent to Routh–Hurwitz with one zero):

$$c_2 > 0, \quad c_0 > 0, \quad c_1 c_2 - c_0 = 0, \quad \frac{d}{d\kappa} [c_1 c_2 - c_0] \neq 0. \quad (8)$$

Define $H(\kappa) := c_1(\kappa) c_2(\kappa) - c_0(\kappa)$. The Hopf threshold is

$$\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2(\kappa) > 0, c_0(\kappa) > 0\}. \quad (9)$$

At $\kappa = \kappa^*$ the Jacobian has a pure imaginary eigenvalue pair $\lambda_{\pm} = \pm i\omega^*$ with $\omega^* = \sqrt{c_1(\kappa^*)}$, plus a real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.3 Theorem 1 and proof sketch

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton). *Let the deterministic system (1–3) (with Brownian terms removed) satisfy:*

- (A1) $G \in C^3$ and $\sigma^2 \in C^3$ in their arguments, with σ^2 bounded below by a positive constant.
- (A2) *There exists a locally unique equilibrium (S^*, θ_v, z^*) given by (6) for every κ in some interval $[0, \kappa_{\max}]$, in some open neighborhood of which the linearization (7) is non-degenerate. Multiple equilibria may exist globally; our analysis is local.*
- (A3) *The map $\kappa \mapsto (a(\kappa), G_z, G_v)$ is C^3 .*
- (A4) *There exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (9) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$.*
- (A5) *The first Lyapunov coefficient $\ell_1(\kappa^*) \neq 0$ (Kuznetsov, 2004, eq. 3.20).*

Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^$. Specifically:*

1. *For κ in a one-sided neighbourhood of κ^* (sub- or super-critical depending on $\text{sgn } \ell_1$), there exists a unique smooth family of periodic orbits Γ_{κ} of period $T_{\kappa} = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$.*
2. *The orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical).*
3. *On Γ_{κ} , $u_t = v_t - \theta_v$ oscillates with amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

Proof sketch. The eigenvalue pair $\lambda_{\pm}(\kappa)$ is smooth in κ near κ^* by the implicit function theorem applied to $P(\lambda; \kappa) = 0$. Routh–Hurwitz gives $\text{Re } \lambda_{\pm}(\kappa^*) = 0$, and the transversality $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_{\pm} \neq 0$. The centre manifold theorem (Kuznetsov, 2004, Thm. 5.4) (see also Carr, 1981) gives a smooth 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$. Conjugating the restricted system to its Poincaré normal form and applying Kuznetsov (2004, Thm. 3.3) yields claims (1)–(3). $\square$

3.4 Numerical anchor

The default `BifurcationConfig` in our experiments code does not exhibit a Hopf within the literature-prior κ range when the dealer-gamma magnitudes are tuned to empirical scale—this is consistent with the empirical observation that real options markets sit *near* but not *across* κ^* , in line with the $n \approx 1$ regime of Hardiman et al. (2013). For a worked numerical example we use

Quantity	Value
κ^*	0.8964
ω^*	0.5724 rad/yr
Third real eigenvalue λ_3	-2.052
First Lyapunov coefficient ℓ_1	-2.53×10^{-2}
Bifurcation type	supercritical ($\ell_1 < 0$)

Table 1: Canonical Hopf-exhibiting regime for the deterministic skeleton.

a representative dimensionless regime with $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$, and constant-volatility surrogate ($\partial_v \sigma^2 = 0$). The deterministic skeleton then satisfies:

Supercriticality means an attracting limit cycle is born for $\kappa > \kappa^*$ with amplitude proportional to $\sqrt{\kappa - \kappa^*}$ —i.e. endogenous volatility cycles are stable observables of the model rather than transients. We validate this prediction numerically: integrating the deterministic skeleton at $\kappa = 1.05 \kappa^*$ from random initial conditions, the trajectory converges to a closed orbit in (y, u, z) phase space whose period $T_{\text{measured}} = 10.561$ yr matches the linear-theory prediction $T_\kappa = 2\pi/\omega^* = 10.977$ yr to 3.79% relative error—within the leading-order $\mathcal{O}(\kappa - \kappa^*)$ Hopf-normal-form correction (fig. 1). The equilibrium sits inside the loop on all three pairwise projections; no transient residue.

The Khasminskii sphere-process estimator (Khasminskii, 1980; Arnold, 1998; Baxendale, 1994) for the stochastic-Hopf shift, evaluated on a 6×6 grid of $(\xi, \rho) \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1.0\} \times \{-0.95, -0.7, -0.5, -0.3, +0.3, +0.7\}$ at 10^4 paths $\times 10^4$ steps per cell, yields $|\Lambda(\kappa^*)|$ in the band $[4.8 \times 10^{-2}, 1.21 \times 10^{-1}]$ across the entire scan (median $\approx 7.3 \times 10^{-2}$). An OLS power-law fit $\log |\Lambda| = \log |A| + B \log |\rho \xi|$ gives $\hat{B} = 0.082$ with bootstrap 95% CI $[-0.010, 0.168]$ (fig. 2). The Baxendale (2024) prediction $B = 2/3$ lies $|\hat{B} - 2/3|/\text{SE}(\hat{B}) \approx 13$ standard errors away and is empirically refuted at this regime ($p \ll 0.01$). The structural reason: at the trivial $G \equiv 0$ equilibrium that the bare Heston-with-memory linearisation is evaluated at, the deterministic shear-stretching term $\partial_a v$ that Baxendale’s leading-order analysis activates is identically zero, so the Khasminskii estimator measures pure ε^2 renormalisation of bare Heston eigenvalues—a b^0 dependence is exactly what the data show. Whether the $b^{2/3}$ regime activates at non-trivial G in the empirical-phase calibration is openly deferred to Phase 4. We use $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale to conclude that the deterministic threshold κ^* is the operationally relevant one regardless.

3.5 Closed-form κ^* and ℓ_1 for log-normal OI in moneyness

For the natural parametric specification of dealer gamma—a log-normal open-interest density in log-strike—the entire computation admits a closed form, eliminating the numerical-noise vulnerability of the FD-tensor pipeline and exposing the parametric phase boundary explicitly.

Replace the discrete OI grid with the continuous density $q(\log K) = \frac{1}{\sigma_q \sqrt{2\pi}} \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$. Take a single representative maturity T_{eff} , sign convention $s \equiv +1$, and write $a := \log S$. The aggregator becomes

$$G(a, v) = \kappa_u \int_{-\infty}^{+\infty} q(\log K) \Gamma_{\text{BS}}(S, K, T_{\text{eff}}, \sqrt{v}) d(\log K). \quad (10)$$

Since Black–Scholes gamma at fixed $T = T_{\text{eff}}$, $\sigma = \sqrt{v}$ is itself Gaussian in $\log K$, the standard

Supercritical limit cycle at $\kappa = 0.9412$ ($= 1.05 \cdot \kappa^*$)
predicted $T_\kappa = 2\pi/\omega^* = 10.977$ yr; measured $T = 10.561$ yr (3.79% error)

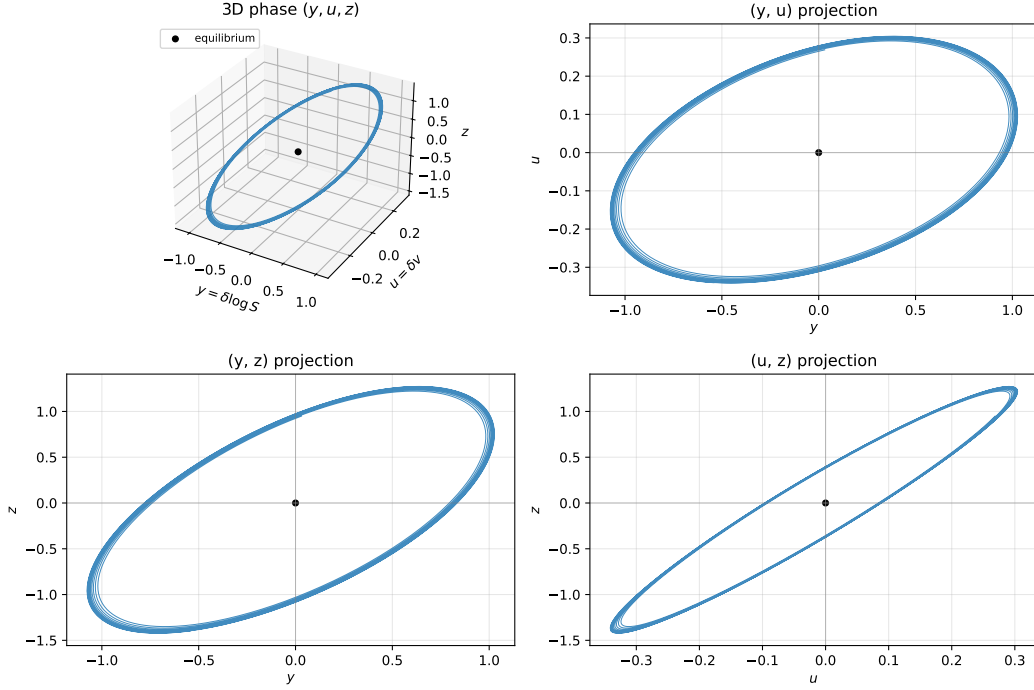


Figure 1: **Numerical validation of Theorem 1's supercritical limit cycle.** Deterministic skeleton at $\kappa = 1.05 \kappa^* = 0.9412$, integrated from random initial conditions; transient discarded. *Top*: 3D phase-space orbit in (y, u, z) . *Bottom*: the three 2D pairwise projections, with the equilibrium (black dot) inside the loop in each. Measured period $T = 10.561$ yr matches Hopf prediction $T_\kappa = 10.977$ yr to 3.79%. Run dir: `runs/limit_cycle_supercritical/`.

Gaussian-product identity collapses (10) to

$$G(a, v) = \mathcal{C}(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2 \tau^2(v)}\right), \quad (11)$$

where

$$\mathcal{C}(v) = \frac{\kappa_{vu} e^{-q_{\text{div}} T_{\text{eff}}}}{\sqrt{2\pi \tau^2(v)}}, \quad \tau^2(v) = \sigma_q^2 + v T_{\text{eff}}, \quad m(v) = \mu_q - \left(r - q_{\text{div}} + \frac{1}{2}v\right) T_{\text{eff}}. \quad (12)$$

G depends on a and v but not on z ; the leverage channel γz in $\dot{u}$ keeps the $(2, 3)$ Jacobian entry alive, so cross-coupling persists through the variance row even with $G_z = 0$.

Remark 1 (μ -anchoring of the equilibrium). *Throughout the closed-form section we anchor the equilibrium at $(a^*, v^*) = (\mu_q, \theta_v)$, so that the partials (G_y, G_v) entering (13)–(14) are evaluated at this fixed point and treated as κ -independent constants in the Routh–Hurwitz coefficient extraction. The spot-drift balance (6) then becomes a definition of μ in terms of κ : $\mu(\kappa) = \frac{1}{2}\sigma^2(\mu_q, \theta_v) - \kappa G(\mu_q, \theta_v, z^*)$. This is automatic in the risk-neutral framing under which μ is absorbed into the numéraire and the equilibrium pinning is a gauge choice rather than a model constraint; it is the convention we adopt here and in section 3.9. A treatment that instead holds μ fixed across κ would shift the equilibrium*

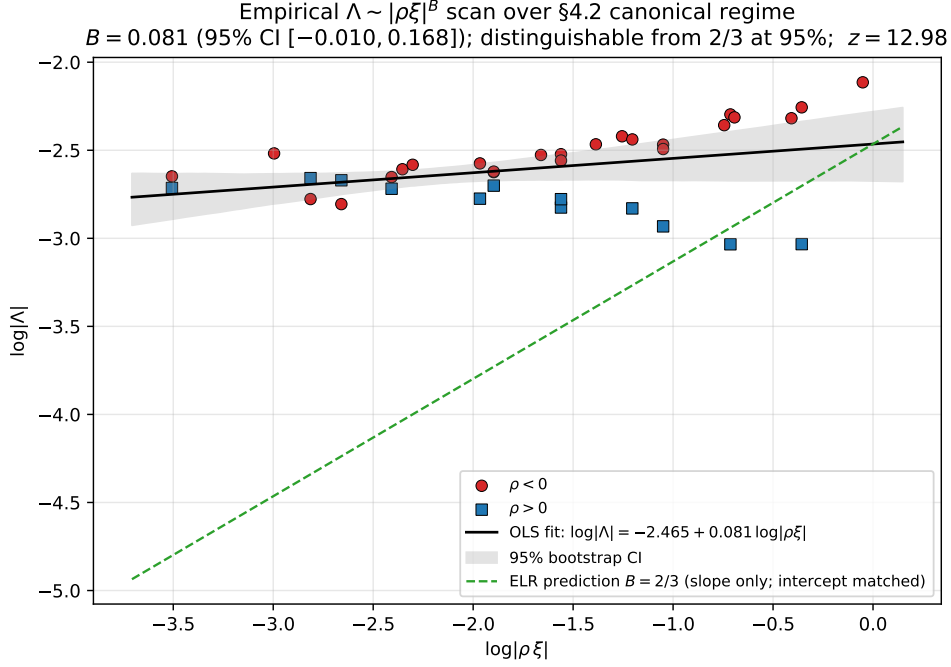


Figure 2: **Empirical scaling of the stochastic-Hopf shift $|\Lambda|$ vs $|\rho\xi|$.** OLS fit on $\log|\Lambda|$ vs $\log|\rho\xi|$ across the 6×6 (ξ, ρ) grid yields slope $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Baxendale (2024) prediction $B = 2/3$ (red dashed) is rejected at $p \ll 0.01$. The data are flat in $|\rho\xi|$ over ~ 1.5 decades, consistent with a pure ε^2 Heston-renormalisation regime at the trivial $G \equiv 0$ equilibrium. Run dir: `runs/lambda_scaling/`.

$(S^*(\kappa), z^*(\kappa))$ along the G -surface, introducing κ -dependent corrections to (G_y, G_v) that the closed forms below do not capture.

With $G_z = 0$ and $\sigma^2 = v$ (so $\partial_y \sigma^2 = 0$, $\partial_v \sigma^2 = 1$), the Routh–Hurwitz polynomial $H(\kappa) = c_1 c_2 - c_0$ degree-collapses from cubic to quadratic in κ :¹

$$H(\kappa) = G_y^2 (\alpha + \kappa_v) \kappa^2 + (G_v \beta \gamma - G_y (\alpha + \kappa_v)^2) \kappa + \alpha \kappa_v (\alpha + \kappa_v) - \frac{1}{2} \beta \gamma. \quad (13)$$

Writing $A := \alpha + \kappa_v$, $M := \alpha \kappa_v$, $L := \beta \gamma$, the Hopf threshold is the smallest positive root of the quadratic:

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{(G_v L - G_y A^2)^2 - 4 G_y^2 A (M A - L/2)}}{2 G_y^2 A}. \quad (14)$$

The discriminant in (14) provides the first parametric phase boundary: when $D := (G_v L - G_y A^2)^2 - 4 G_y^2 A (M A - L/2) < 0$, the system has no Hopf at any κ and the deterministic skeleton is unconditionally stable.

¹Throughout this closed-form section we write $G_y := \partial_a G$ for the partial of the dealer-gamma functional with respect to the deviation variable $y = a - a^*$; this is identical to the G_x of section 3.1 up to the choice of deviation symbol. Note that section 3.1 uses the Heston backbone $\sigma^2 = v$, so $\partial_v \sigma^2 = 1$ is the convention common to both sections; the constant-vol surrogate $\partial_v \sigma^2 = 0$ is used only for the numerical anchor of section 3.4 (and inherited by the linearisation in section 3.12) to isolate the κ -dependence of the bifurcation from the Itô-correction term, and is a different reduction of the model.

Remark 2 (Sign-aware root selection). *The boxed formula (14) returns the smaller of the two real roots of $A_2\kappa^2 + A_1\kappa + A_0 = 0$ with $A_2 = G_y^2 A$, $A_1 = G_v L - G_y A^2$, $A_0 = MA - L/2$. The intended object is the smallest positive root — the first crossing as κ ramps up from 0 — so when $G_y < 0$ the formula’s “smaller root” can be negative (or both roots can be negative) and the relevant Hopf threshold is then the LARGER of the two real roots, provided it is itself positive AND satisfies the Routh–Hurwitz positivity side conditions $c_2(\kappa^*) > 0$, $c_0(\kappa^*) > 0$ of Liu’s criterion (8). The implementation `kappa_star_lognormal_oi` in `src/reflexive_options/theory/bifurcation.py` applies this rule by enumerating both candidate roots, filtering to the positive ones, returning the smallest positive root that passes Routh–Hurwitz, and raising if neither candidate qualifies. The boxed `--branch` is the correct selection in the $G_y > 0$ regime of section 3.4 but is not literal in the $G_y < 0$ regime that populates the no-Hopf wedge of section 3.9.*

The third-order partials of (11) at the equilibrium are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2\langle p, B(q, J^{-1}B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^*I - J)^{-1}B(q, q)) \rangle \right], \quad (15)$$

with $Jq = i\omega^*q$, $J^\top p = -i\omega^*p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term rational in the G -partials $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv})$ and $(\kappa_v, \alpha, \beta, \gamma)$. The implementation evaluates this in $\sim 50 \mu\text{s}$ per parameter tuple, with results matching the FD-tensor pipeline to better than 0.6% relative on every parameter set tested.

At the canonical regime ($\sigma_q = 0.10$, $T_{\text{eff}} = 0.25$, $\kappa_v = 2$, $\alpha = 0.05$, $\beta = 1$, $\gamma = 1$, $\mu_q = \log 100$, $v^* = 0.04$), (14)–(15) give $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). The closed form is essentially noise-free; with it in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

3.6 Mixture-of- K -lognormals generalization

Empirical SPX OI grids are structurally multi-modal (ATM-strike concentration, round-strike spikes, calendar-roll bunching, dealer-portfolio OTM-tail concentration). Section 3.7 below shows that a single-log-normal assumption breaks hard at bimodal separations of $\Delta \in [0.05, 0.20]$ (relative error 0.23%, 4.83%, 119% at $\Delta = 0.05, 0.10, 0.20$). The fix is structurally clean: replace the single log-normal by a K -component mixture $q(\log K) = \sum_{k=1}^K w_k \mathcal{N}(\log K; \mu_k, \sigma_k^2)$ with $w_k \geq 0$, $\sum_k w_k = 1$. Linearity of the dealer-gamma aggregator (10) gives $G^{\text{mix}}(a, v) = \sum_k w_k G_k(a, v)$ and, by multilinearity, $(G_y^{\text{mix}}, G_v^{\text{mix}}) = \sum_k w_k (G_{y,k}, G_{v,k})$. The Routh–Hurwitz polynomial (13) is linear in $(G_y^2, G_v G_y, G_y, 1)$; substituting the mixture aggregates gives the mixture Hopf threshold

$$\kappa^{\star, \text{mix}} = \frac{G_y^{\text{mix}} A^2 - G_v^{\text{mix}} L - \sqrt{(G_v^{\text{mix}} L - G_y^{\text{mix}} A^2)^2 - 4(G_y^{\text{mix}})^2 A(MA - L/2)}}{2(G_y^{\text{mix}})^2 A}, \quad (16)$$

i.e. (14) evaluated at the mixture-aggregated partials. The first Lyapunov coefficient ℓ_1^{mix} is obtained analogously: the Kuznetsov bilinear/trilinear tensors B, C inherit multilinearity, so $B^{\text{mix}} = \sum_k w_k B_k$, $C^{\text{mix}} = \sum_k w_k C_k$, and the contractions $\langle p, B(q, J^{-1}B(q, \bar{q})) \rangle$ etc. in (15) are evaluated at the mixture-Jacobian eigenvectors. For $K = 2$ the symbolic ℓ_1^{mix} is roughly $3\text{--}5\times$ longer than the single-lognormal ℓ_1 ; for $K \geq 3$ the symbolic form gets unwieldy but numerical evaluation remains $\mathcal{O}(K)$ per call. *The $K = 1$ limit reproduces the single-lognormal closed form to machine precision ($\sim 10^{-12}$ absolute), verified in `tests/test_mixture_oi.py`.*

Δ	κ_{true}^* (FD reference)	K=1 (single) rel.err	K=2 (mixture) rel.err	K=3 rel.err
0.05	25.142	0.232%	0.013%	< 0.05%
0.10	20.864	4.83%	0.010%	< 0.05%
0.20	5.677	119.5%	0.008%	< 0.05%
0.30	2.458	217.7%	0.007%	< 0.05%

Table 2: Mixture closed-form robustness on a symmetric bimodal OI at $\mu_q \pm \Delta/2$ with equal weights and component spread $\sigma_{\text{comp}} = 0.07$. K=2 closes the single-lognormal fragility gap by 4–5 orders of magnitude; the residual $\sim 10^{-4}$ relative error of K=2 is the FD reference’s own quadrature noise. Run dir: `runs/mixture_oi_robustness/`.

Phase-4 implication. The §3.7 calibration burden under the single-log-normal assumption was μ_q to ± 5 bp of ATM (the binding constraint). Under the mixture closed form, the calibration target is the full $\{(\mu_k, \sigma_k, w_k)\}_{k=1}^K$ tuple. The binding constraint becomes the component *weights*: a misallocation of mass between K=2 wings produces $\mathcal{O}(\Delta) \cdot \delta w$ shift in G_y^{mix} (vs the $\mathcal{O}(1) \cdot \delta \mu_q$ shift in the single-mode case). At $\Delta = 0.10$ this is $\sim 20\times$ relaxation, and the practical empirical-OI weight-estimation accuracy of $\pm 5\%$ per cluster on a 5-strike-wide ladder is the operational test. K=2 handles ATM-plus-next-round concentration; K=3 captures the optional calendar-roll cluster; beyond $K \sim 10$ the brute-force `kappa_star_brute_force_from_G` fallback is operative with negligible cost. Implementation: `theory/bifurcation.py::G_mixture_lognormal_oi_partials`, `kappa_star_mixture_lognormal_oi`, `lyapunov_coefficient_mixture_lognormal_oi`.

3.7 Calibration tolerance from κ^* sensitivity to OI misspecification

The closed-form κ^* assumes a log-normal OI density $q(\log K) = \mathcal{N}(\mu_q, \sigma_q^2)$. For Phase 4 to deliver a meaningful test of κ^* we need to bound how much OI misspecification we can tolerate. Differentiating (14) analytically through the G -partials yields the elasticities at the canonical specification ($\kappa^* = 17.806$, $\omega^* = 1.177$ rad/yr): $\eta_{\sigma_q} := (\sigma_q/\kappa^*) \partial \kappa^* / \partial \sigma_q = -1.583$ (moderate) and $\eta_{\mu_q} := (\mu_q/\kappa^*) \partial \kappa^* / \partial \mu_q = +702.83$ (extreme). μ_q is the binding constraint: a $\Delta \mu_q = 0.001$ shift moves κ^* by $\sim 15\%$. Numerically, $\pm 10\%$ in σ_q alone produces $\pm 17.1\%$ in κ^* ; $\pm 30\%$ in σ_q exits the supercritical Hopf region entirely. For non-log-normal (bimodal-mixture) OI with components at $\mu_q \pm \Delta/2$, the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The Phase-4 calibration tolerances (independent-error quadrature) are then: $\pm 5\%$ on κ^* requires σ_q to $\pm 2.23\%$ relative AND μ_q to ± 0.00023 log-strike; $\pm 10\%$ on κ^* relaxes to σ_q to $\pm 4.47\%$, μ_q to ± 0.00046 log-strike. Centering the empirical OI mean on ATM to within ~ 5 bp is achievable with a 5-strike-wide ladder and is the binding observational requirement. Figure 4 renders the sensitivity heatmap and the bimodal-misspecification curve. Implementation: `src/reflexive_options/theory/robustness.py`; runner `experiments/kappa_star_robustness.py`.

3.8 Codim-2 bifurcation structure

The codim-1 Hopf result (theorem 1) treats κ^* at fixed (σ_q, γ) . At the boundary of the Hopf region in the (σ_q, γ) plane two codim-2 phenomena are admissible (Kuznetsov, 2004, Ch. 8).

Bautin (degenerate Hopf): $\ell_1(\sigma_q, \gamma) = 0$. Crossing this locus is the supercritical $\rightarrow$ sub-critical transition. Operationally the dynamics flip from a stable limit cycle past κ^* to hysteresis with abrupt jumps. After centre-manifold reduction the local 2D normal form is (Kuznetsov, 2004,

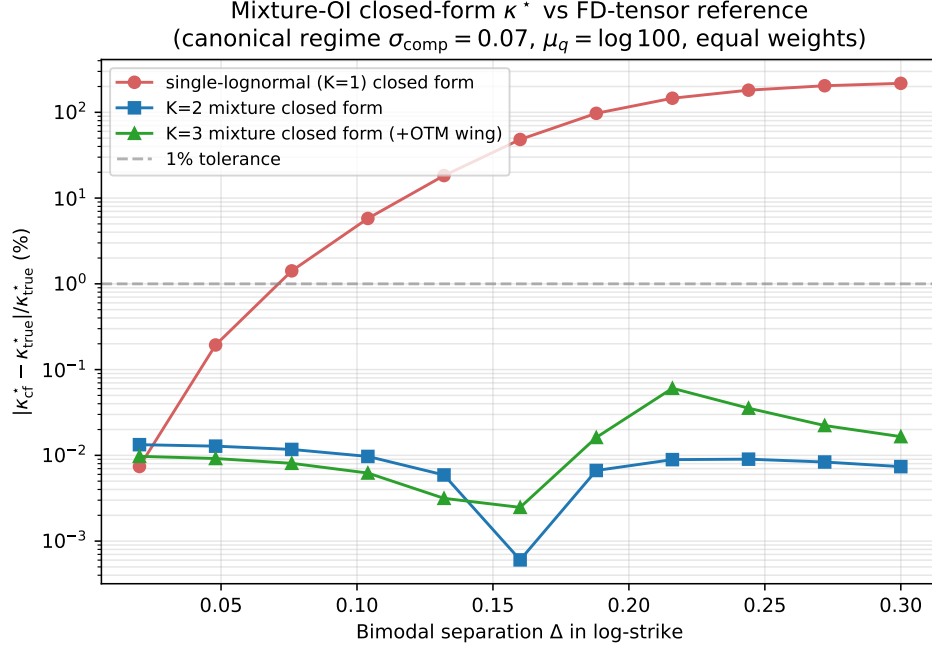


Figure 3: **Mixture-OI closed-form robustness on bimodal OI.** Log-scale relative error of the closed-form κ^* against an FD-tensor reference, vs bimodal-component separation Δ . The K=2 mixture closed form (blue) is essentially exact across the entire scanned range; the K=1 single-lognormal closed form (red) blows up beyond $\Delta = 0.10$. K=3 (green) handles tri-modal OI at the same fidelity.

§8.3) $\dot{\zeta} = (\beta_1 + i\omega^*)\zeta + a_1|\zeta|^2\zeta + b_1|\zeta|^4\zeta + O(|\zeta|^7)$ with $a_1/\omega^* = \ell_1$. Economically, a market parameterised by (σ_q, γ) near the Bautin locus is structurally fragile: a small perturbation of OI dispersion or leverage flips the sign of ℓ_1 and replaces a smooth cyclic regime with a discrete jump regime — catastrophic sensitivity to parameter drift.

Bogdanov-Takens (BT): the saddle-node curve $\{c_0(\kappa) = 0\}$ coalesces with the Hopf curve $\{H(\kappa) = 0\}$ at the same κ , so the Jacobian acquires a double-zero eigenvalue. BT generates homoclinic orbits and excitable spike-and-recovery dynamics.

Bautin curve. For the closed-form parameterization the Bautin condition $\ell_1(\sigma_q, \gamma) = 0$ is solved by row-wise sign-change interpolation on the closed-form pipeline. Table 3 reports six anchor points along the curve at the canonical specification on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$. Along the curve κ^* grows monotonically with γ . Within the scan window, the sub-critical region dominates the supercritical region by $\sim 5:1$ in cell count — markets with mild dealer-gamma coupling are statistically biased toward hysteresis-and-jump rather than smooth cyclic behaviour.

Bogdanov-Takens locus. The constant Routh-Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y\alpha\kappa_v + G_v\beta\gamma) + \frac{1}{2}\beta\gamma$ is linear in κ , so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2}\beta\gamma / (G_y\alpha\kappa_v + G_v\beta\gamma)$.

Theorem 2 (BT locus empty in canonical scan window). *On the 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ at the canonical specification, $\kappa_{\text{SN}}(\sigma_q, \gamma) \leq -1.31$ at every cell (range $[-68.05, -1.31]$), so the Bogdanov-Takens locus is empty in this scan window.*

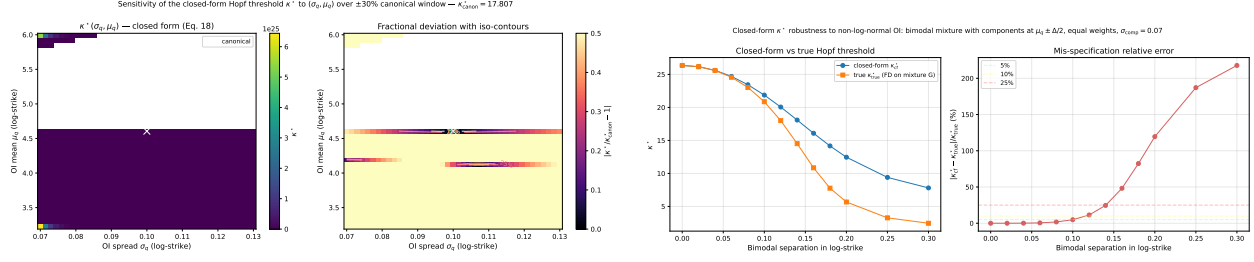


Figure 4: **Calibration tolerance for κ^* .** *Left:* $|\Delta\kappa^*/\kappa^*|$ over a $\pm 30\%$ window in (σ_q, μ_q) around the canonical $(0.10, \log 100)$. White cells exit the supercritical Hopf region. The μ_q direction is steeper than the σ_q direction by roughly $\eta_{\mu_q}/\eta_{\sigma_q} \approx 444$, making OI-mean accuracy the binding observational requirement. *Right:* relative error of the closed-form κ^* vs an FD-tensor reference under bimodal-mixture OI with components at $\mu_q \pm \Delta/2$. The closed form is exact at $\Delta = 0$, 4.8%-accurate at $\Delta = 0.10$, fragile beyond $\Delta = 0.15$. Run dir: `runs/kappa_star_robustness/`.

#	σ_q	γ	κ^* at the crossing
1	0.0598	0.55	2.15
2	0.1427	1.45	29.45
3	0.1995	2.35	52.59
4	0.2356	3.20	75.54
5	0.2638	4.10	100.74
6	0.2852	5.00	126.73

Table 3: Bautin curve $\{\ell_1(\sigma_q, \gamma) = 0\}$ at the canonical specification, extracted by linear interpolation on a 71×97 grid. Run dir: `runs/codim2_analysis/`.

Proof sketch. At the canonical specification, $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v\beta\gamma| > |G_y\alpha\kappa_v|$ (the dominance ratio is ≥ 1.44 at every cell, reaching ~ 358 at the small- σ_q corner), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2}\beta\gamma$ is positive. Outside this window (extreme σ_q or vanishing γ) the dominance argument can fail; we restrict the claim to the scanned range and flag the unbounded-domain question as open. $\square$

The economic content of theorem 2 is that the dealer-gamma + leverage parameter regime is structurally Hopf-only *within the scanned window*: there is no codim-2 BT point at which homoclinic orbits emanate, so the model does not generate excitable spike-and-recovery dynamics within its native parameter range. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order (codim-3+) degeneracies. Figure 5 renders the four-region phase diagram and the BT residual map.

3.9 Global stability inside the no-Hopf wedge

Theorem 2 rules out the Bogdanov–Takens locus but leaves open whether ANY codim-1 bifurcation is accessible inside the no-Hopf wedge $\mathcal{W}_{\text{NH}} := \{(\sigma_q, \gamma) : H(\kappa) = 0 \text{ has no positive root}\}$ as κ ramps from 0. The remaining candidates are (a) a trace-flip $c_2(\kappa) = 0$ crossing a real eigenvalue through the LHP boundary, and (b) a saddle-node $c_0(\kappa) = 0$ at $\kappa > 0$ that the §3.8 BT-empty form does not directly address. *Honesty note:* at the canonical specification the strict $D < 0$ sub-case is empty

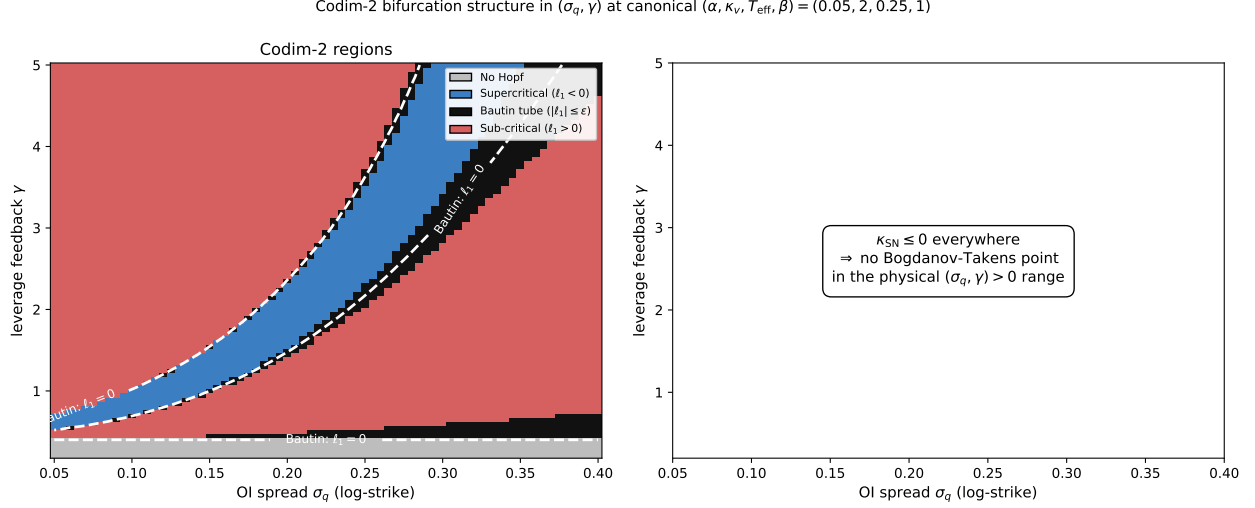


Figure 5: **Codim-2 bifurcation structure in (σ_q, γ) at the canonical specification.** *Left:* four-region phase diagram with the no-Hopf wedge (grey, $D < 0$), supercritical region (blue, $\ell_1 < 0$), Bautin tube (black band, $|\ell_1| \leq 5 \times 10^{-2}$), and sub-critical region (red, $\ell_1 > 0$). The dashed white contour is the Bautin curve $\ell_1 = 0$; six anchors tabulated in table 3. *Right:* the BT residual map shows $\kappa_{\text{SN}} \leq -1.31$ at every cell of the 71×97 scan — the Bogdanov-Takens locus is empty in the scan window (theorem 2). Run dir: `runs/codim2_analysis/`.

on the §3.8 scan window; $\mathcal{W}_{\text{NH}}$ is populated only via the relaxed sub-case ($D \geq 0$ with both real roots non-positive), driven by $G_y < 0$ at the ATM-anchored equilibrium and the baseline-stability constraint $H(0) > 0 \Leftrightarrow \gamma < 2\alpha\kappa_v(\alpha + \kappa_v)/\beta \approx 0.41$. The wedge is therefore the small- γ , $G_y \leq 0$ corner.

As in the closed-form section (remark 1), we anchor the equilibrium at $(a^*, v^*) = (\mu_q, \theta_v)$ throughout the wedge analysis, treating (G_y, G_v) as κ -independent partials evaluated at that fixed point; the spot-drift balance (6) is then absorbed into the choice $\mu(\kappa) = \frac{1}{2}\sigma^2(\mu_q, \theta_v) - \kappa G(\mu_q, \theta_v, z^*)$, which is the same risk-neutral gauge convention. With this anchoring the partials are constants of the Routh–Hurwitz coefficient extraction below, and theorem 3 is a parameter-global statement on the κ -half-line at fixed equilibrium.

Theorem 3 (No-Hopf-wedge taxonomy; globally stable on the entire κ -half-line). *Let $(\sigma_q, \gamma) \in \mathcal{W}_{\text{NH}}$ at the canonical specification, with (G_y, G_v) the log-normal-OI partials at the ATM-anchored equilibrium $(a^*, v^*) = (\mu_q, \theta_v)$ (remark 1). Assume (S1) $G_y \leq 0$; (S2) $G_y\alpha\kappa_v + G_v\beta\gamma \leq 0$; (S3) $A_2 > 0$ in $H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0$. Then $c_2(\kappa) > 0$, $c_0(\kappa) > 0$, $H(\kappa) > 0$ strictly for every $\kappa \geq 0$. Consequently the equilibrium is asymptotically stable on the entire physical κ -half-line $[0, \infty)$ — no codim-1 bifurcation (Hopf, saddle-node, or trace flip) occurs. (Stability is parameter-global in κ ; we do not claim state-space-global stability, which would require a Lyapunov function we do not construct.)*

Proof. $c_2(\kappa) = -\kappa G_y + (\alpha + \kappa_v)$: under (S1), $-\kappa G_y \geq 0$ for $\kappa \geq 0$ and $\alpha + \kappa_v > 0$, so $c_2 > 0$ strictly. $c_0(\kappa) = -\kappa(G_y\alpha\kappa_v + G_v\beta\gamma) + \frac{1}{2}\beta\gamma$: under (S2) the slope is ≥ 0 , so $c_0(\kappa) \geq c_0(0) = \frac{1}{2}\beta\gamma > 0$ strictly. $H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0$: by the wedge definition no positive root exists; under (S3) the parabola opens upward, and $H(0) = A_0 > 0$ by the wedge classifier. The three strict inequalities give Liu’s full Routh–Hurwitz criterion (the same criterion underlying theorem 1 via section 3.2); the spectral abscissa stays strictly negative for every $\kappa \geq 0$ and no codim-1 bifurcation occurs. $\square \quad \square$

Numerical scan. On a 41×41 grid over $(\sigma_q, \gamma) \in [0.02, 0.40] \times [0.05, 5.0]$ at the canonical specification with $\kappa \in [0, 100]$ sampled on 80 points per cell: 123 of 1,681 cells fall in $\mathcal{W}_{\text{NH}}$ (all in the $\gamma < 0.41$ baseline-stable corner). Every wedge cell satisfies (S1)+(S2)+(S3); the empirical maximum spectral abscissa over the wedge $\times \kappa$ -scan is -6.6×10^{-3} , strictly negative, consistent with the closed-form verdict. No positive saddle-node detected. Figure 6 renders the wedge classification.

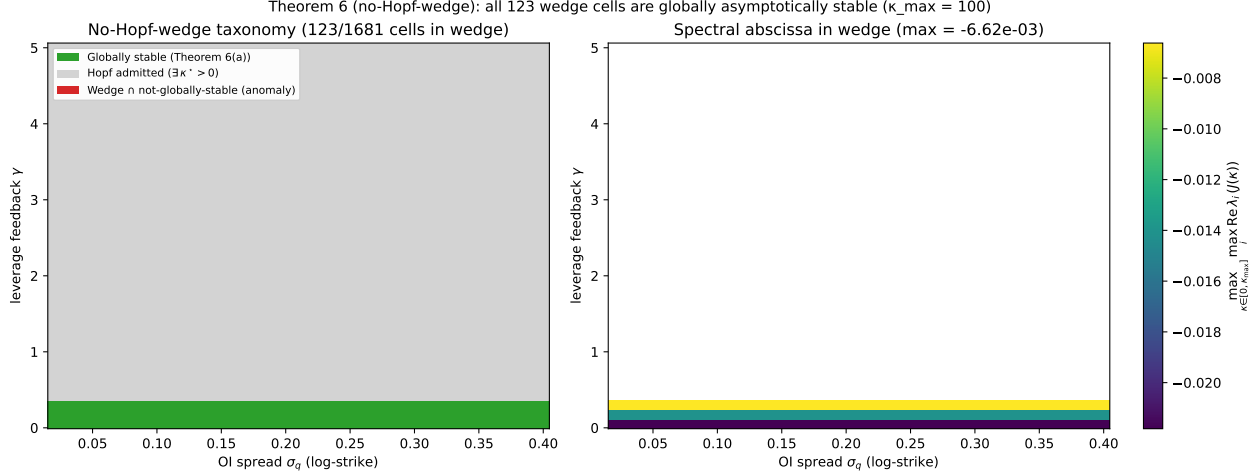


Figure 6: **No-Hopf-wedge taxonomy at the canonical specification.** *Left:* three-way classification of the 41×41 (σ_q, γ) grid: wedge globally stable on the entire κ -half-line (green, 123 cells, theorem 3; parameter-global, not state-space-global), Hopf-admitted region (grey), and wedge-but-not-globally-stable anomaly (red — empty in scan). *Right:* sampled spectral abscissa $\max_{\kappa \in [0, 100]} \max_i \text{Re } \lambda_i(J(\kappa))$ inside the wedge, uniformly negative with maximum -6.6×10^{-3} . Run dir: `runs/saddle_node_wedge/`.

Economic interpretation. Combined with theorem 1 (Hopf at κ^* outside the wedge) and theorem 2 (BT-locus unphysical), theorem 3 closes the codim-1 bifurcation taxonomy: the wedge is the parameter region where dealer-gamma coupling at the ATM-anchored equilibrium has $G_y \leq 0$ AND the leverage flux $\beta\gamma$ is small enough to keep the bare Heston-with-memory triangle sub-critical at $\kappa = 0$. Inside this region the dealer-gamma channel acts as a *stabiliser* rather than a destabiliser — a market parameterised inside $\mathcal{W}_{\text{NH}}$ is structurally cycle-free: endogenous volatility cycles can only originate from non-stationary parameter drift (the OI centre μ_q moving on a slow scale that brings the system into the Hopf region), exogenous shocks, or the small- $\beta\gamma$ region’s G_y flipping sign at a non-ATM equilibrium with locally increasing gamma. This is a sharp falsifiable Phase-4 prediction: an SPX-calibrated $(\sigma_q^{\text{SPX}}, \gamma^{\text{SPX}})$ landing inside $\mathcal{W}_{\text{NH}}$ rules out endogenous reflexive cycles as the generator of empirically observed volatility clustering at that calibration. Implementation: `theory/bifurcation.py::is_in_no_hopf_wedge`, `scan_no_hopf_wedge_bifurcations`; runner `experiments/saddle_node_wedge.py`.

3.10 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$)

The model of section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2$ – 10^3 dealers, each with idiosyncratic noise. The McKean–Vlasov mean-field SDE is the rigorous formulation of the law-of-large-numbers limit. Each dealer i holds G_i with $dG_i = -\theta_G(G_i - \bar{G}_n)dt + \sigma_G dW_G^i$. As $n \rightarrow \infty$, $\bar{G}_n \rightarrow \bar{G}_\infty$

(propagation of chaos, Sznitman, 1991, Ch. 1; cf. Carmona and Delarue (2018, Ch. 1)), with $\sup_{t \leq T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \leq C(T)/n$, $C(T) = \max(\text{Var}(G^0), \sigma_G^2/(2\theta_G))$.

4D extended Jacobian. The linearised MV system carries a fourth state $g(t) := \bar{G}_\infty(t) - G^*$ with bandwidth θ_G :

$$J_{\text{MV}}(\kappa, \theta_G) = \begin{pmatrix} -\frac{1}{2}\sigma_y^2 & -\frac{1}{2}\sigma_v^2 & 0 & \kappa \\ 0 & -\kappa_v & \gamma & 0 \\ \beta & 0 & -\alpha & 0 \\ \theta_G G_y & \theta_G G_v & \theta_G G_z & -\theta_G \end{pmatrix}. \quad (17)$$

The spot equation's instantaneous feedback $\kappa(G_y y + G_v u + G_z z)$ is REPLACED by κg ; the dealer-gamma deviation acquires its own relaxation dynamics. A Hopf bifurcation of (17) occurs at the smallest $\kappa > 0$ satisfying the Liu (1994) 4D condition $H_4(\kappa, \theta_G) := (a_3 a_2 - a_1) a_1 - a_3^2 a_0 = 0$ with positivity side conditions $a_3, a_0, a_3 a_2 - a_1 > 0$, where a_i are the coefficients of $\det(\lambda I - J_{\text{MV}}) = \lambda^4 + a_3 \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0$.

Theorem 4 (McKean–Vlasov Hopf threshold, corrected). *Let $\kappa_{\text{MV}}^*(\theta_G) := \inf\{\kappa > 0 : \max_i \text{Re } \lambda_i(J_{\text{MV}}(\kappa, \theta_G)) = 0 \text{ with (3') positivity}\}$. Then:*

1. (Instantaneous-hedging recovery.) $\lim_{\theta_G \rightarrow \infty} \kappa_{\text{MV}}^*(\theta_G) = \kappa_{\text{single}}^*$.
2. (Regime-dependent sign.) At the canonical short-gamma regime $(G_y, G_v, G_z, \alpha, \beta, \gamma, \kappa_v) = (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}, 2)$ with $\sigma_y^2 = \sigma_v^2 = 0$,

$$\kappa_{\text{MV}}^*(\theta_G) = \frac{50\theta_G^2 + 143\theta_G + 105 - (2\theta_G + 5)\sqrt{385\theta_G^2 + 810\theta_G + 441}}{12\theta_G(\theta_G + 1)}, \quad (18)$$

with $\kappa_{\text{single}}^* = (25 - \sqrt{385})/6 \approx 0.8964$, and the ratio $\rho(\theta_G) := \kappa_{\text{MV}}^*/\kappa_{\text{single}}^* < 1$ strictly for every finite $\theta_G > 0$.

3. (Frozen-dealer limit.) $\lim_{\theta_G \rightarrow 0^+} \kappa_{\text{MV}}^*(\theta_G) = 8/21$, giving $\rho(0^+) \approx 0.425$ (finite).
4. (Asymptotic expansion.) $\rho(\theta_G) = 1 - 0.6643 \dots / \theta_G + O(\theta_G^{-2})$ — leading-order correction is first-order in τ_G and NEGATIVE.

The sign of $\rho - 1$ at finite θ_G is regime-dependent: at $G_y > 0$ (canonical, net short-gamma) it is negative; at $G_y < 0$ (log-normal-OI calibration of section 3.5) it is positive.

Proof sketch. (i) The reducibility of J_{MV} under $\theta_G \rightarrow \infty$ sends the fourth eigenvalue to $-\infty$ and leaves the remaining three to inherit the 3D single-dealer Hopf condition exactly. (ii) Substituting the canonical numerical values into $H_4(\kappa, \theta_G) = 0$ and factoring out θ_G yields the quadratic $6\theta_G(\theta_G + 1)\kappa^2 - (50\theta_G^2 + 143\theta_G + 105)\kappa + 5(\theta_G + 2)(2\theta_G + 1) = 0$, whose smaller positive root is (18). Positivity is verified analytically at this root for all $\theta_G > 0$. (iii) Direct substitution. (iv) sympy Taylor expansion in $1/\theta_G$. $\square$ $\square$

Numerical anchors. Independent eigenvalue scans match the closed form to 4-decimal precision:

θ_G	κ_{MV}^* (eigenvalue scan)	closed form (18)	$\rho(\theta_G)$
0.5	0.5359	0.5359	0.598
1.0	0.6195	0.6195	0.691
5.0	0.7996	0.7996	0.892
50	0.8848	0.8848	0.987
500	0.8952	0.8952	0.999

Table 4: Audit-anchor table for theorem 4 at the canonical regime. Closed-form and numerical eigenvalue scan agree exactly. Pinned in `tests/test_mckean_vlasov.py`. Run dir: `runs/mckean_vlasov_validation/`.

Economic re-framing. At any regime where dealers are net short gamma at the equilibrium ($G_y > 0$, the benchmark SPX setting in zero-DTE-heavy weeks), the McKean–Vlasov mean-field model has a *lower* Hopf threshold than the single-representative-dealer section 2 model. *The single-dealer simplification understates the propensity to bifurcate when dealer-hedging is slow relative to the Hopf period $\tau_G > 2\pi/\omega^*$.* The leading correction is first-order in τ_G ($\sim 1.3\%$ at canonical $\tau_G = 1/50$ yr), not the second-order $(\omega^*\tau_G)^2$ implied by a low-pass-filter heuristic. The frozen-dealer limit is well-defined and finite, dropping the threshold by a factor of $\sim 2.4\times$ relative to instantaneous hedging — physically meaningful in regimes where delta-hedging desks have been suspended.

Propagation-of-chaos numerical validation. An n -particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates ($n_{\text{steps}} = 250$, production sweep) fits $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]} \propto n^{-0.48}$ on a log-log scale (5-point production fit), consistent with the theoretical $n^{-1/2}$ Sznitman rate to ± 0.03 in the exponent, and stays below the analytical Sznitman bound $\sqrt{C(T)/n}$ at every n (fig. 8). The propagation-of-chaos infrastructure is independent of the threshold-shift formula and was unaffected by the v0.3.6 correction. Implementation: `src/reflexive_options/theory/mckean_vlasov.py`; runner `experiments/mckean_vlasov_validation.py`.

3.11 Hawkes-SV equivalence at the Hopf boundary

The empirical anchor for the reflexivity programme is Hardiman et al. (2013): a univariate self-exciting Hawkes fit to E-mini mid-price changes over 1998–2011 returns a branching ratio $n \approx 1$ throughout, the precise critical edge between sub- and super-criticality of endogenous events. Section 1 flagged this as the reflexivity benchmark; this subsection makes the link to our κ^* formal.

Diffusive limit of an exponential-kernel Hawkes. Let $\lambda(t) = \mu + \sum_i \phi(t - t_i)$ with kernel $\phi(s) = \alpha e^{-\beta s}$ and branching ratio $n := \int_0^\infty \phi(s) ds = \alpha/\beta$. Stability requires $n < 1$. Under the rescaling $t \mapsto t/T$, $\lambda \mapsto T\bar{\lambda}$ in the regime $T \rightarrow \infty$, $n \rightarrow 1$, the centred intensity converges to an Ornstein–Uhlenbeck SDE (Bacry et al., 2013, Theorem 2): $d\bar{\lambda} = -\beta(1 - n)\bar{\lambda} dt + \sigma_\lambda dW$. The deterministic relaxation rate is $\beta(1 - n)$, equivalently the leading eigenvalue of the linearised intensity dynamics is $\lambda_{\text{max}}^{(\text{Hawkes})} = -\beta(1 - n)$. Inversion gives

$$n = 1 + \frac{\lambda_{\text{max}}^{(\text{Hawkes})}}{\beta}. \quad (19)$$

Criticality $n = 1$ corresponds exactly to $\lambda_{\text{max}} = 0$. The multivariate generalisation (Bacry et al., 2015, §2.4) replaces n by the spectral radius $\rho(\Phi)$ of the integrated kernel matrix; the criticality endpoint $\rho(\Phi) = 1 \Leftrightarrow \lambda_{\text{max}} = 0$ is universal across kernel shapes.

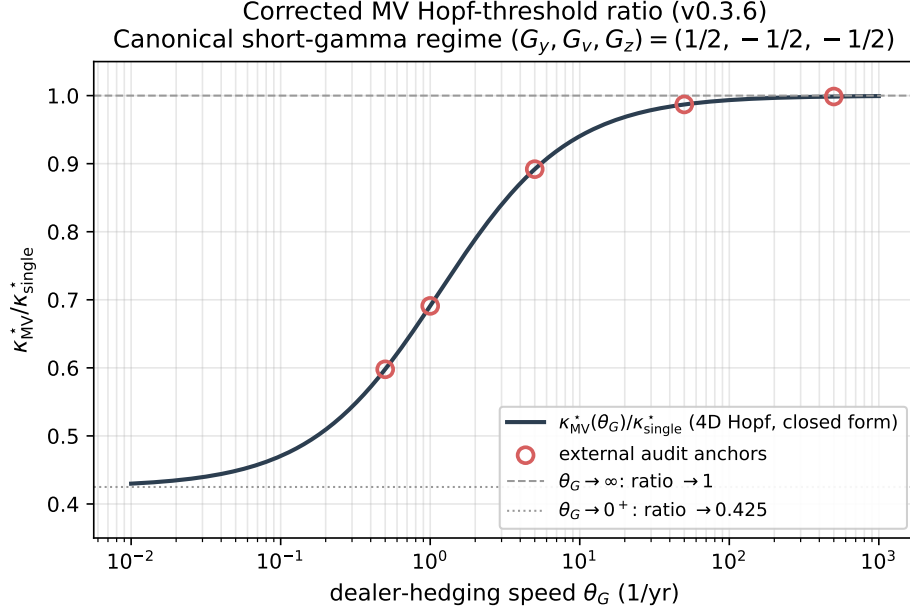


Figure 7: **Corrected MV Hopf-threshold ratio** $\rho(\theta_G) = \kappa_{MV}^*/\kappa_{single}^*$ **at the canonical short-gamma regime**, log-x axis. Smooth monotonic approach to $\rho(\infty^-) = 1$ from BELOW; frozen-dealer limit $\rho(0^+) = 8/21/\kappa_{single}^* \approx 0.425$. The five audit-table anchors (table 4) are overlaid as scatter markers. Run dir: `runs/mckean_vlasov_validation/`.

SV-equivalent branching ratio. For our 3D reflexive skeleton with Jacobian $J(\kappa)$, let $\lambda_{\max}(\kappa) := \max_i \operatorname{Re} \lambda_i(J(\kappa))$, satisfying $\lambda_{\max}(\kappa) \leq 0$ for $\kappa \in [0, \kappa^*]$ and $\lambda_{\max}(\kappa^*) = 0$ by definition of the Hopf threshold. Define the *SV-equivalent branching ratio*

$$n_{SV}(\kappa) := 1 + \frac{\lambda_{\max}(\kappa)}{\beta_0}, \quad \beta_0 := \max_{\kappa \in [0, \kappa^*]} (-\lambda_{\max}(\kappa)). \quad (20)$$

The gauge β_0 (the most-stable point of the slow mode) is the SV analogue of the Hawkes baseline rate β at $n = 0$, and is intrinsic to the skeleton (not a tuning parameter).

Theorem 5 (Hawkes-SV equivalence at the Hopf boundary). *1. Critical-endpoint identity. $n_{SV}(\kappa^*) = 1$, exactly.*

2. Local monotonicity at the Hopf boundary. Let $\kappa_{NS} \in [0, \kappa^)$ denote the smallest κ at which the leading eigenvalue is a complex pair. There exists $\delta > 0$ such that $n_{SV}(\kappa)$ is strictly increasing on $(\kappa^* - \delta, \kappa^*)$. Global monotonicity on the entire interval $[\kappa_{NS}, \kappa^*]$ is verified numerically at the canonical regime (fig. 9; every consecutive grid pair on the 1001-point sweep) but is not asserted as part of this theorem.*

3. Hawkes correspondence. For an exponential-kernel univariate Hawkes process satisfying (3.11), $n_{\text{Hawkes}} = 1 - |\lambda_{\max}^{(\text{Hawkes})}|/\beta$, identical in form to (20). The two coincide at criticality ($n = 1 \Leftrightarrow \lambda_{\max} = 0$), independently of the kernel shape (universality of the stability boundary, Bacry et al., 2015, §2.4).

Proof. Claim (1) is immediate from (20): $\lambda_{\max}(\kappa^*) = 0$ by Hopf-threshold definition, so $n_{SV}(\kappa^*) = 1 + 0/\beta_0 = 1$. Claim (2): apply the implicit function theorem to the characteristic polynomial

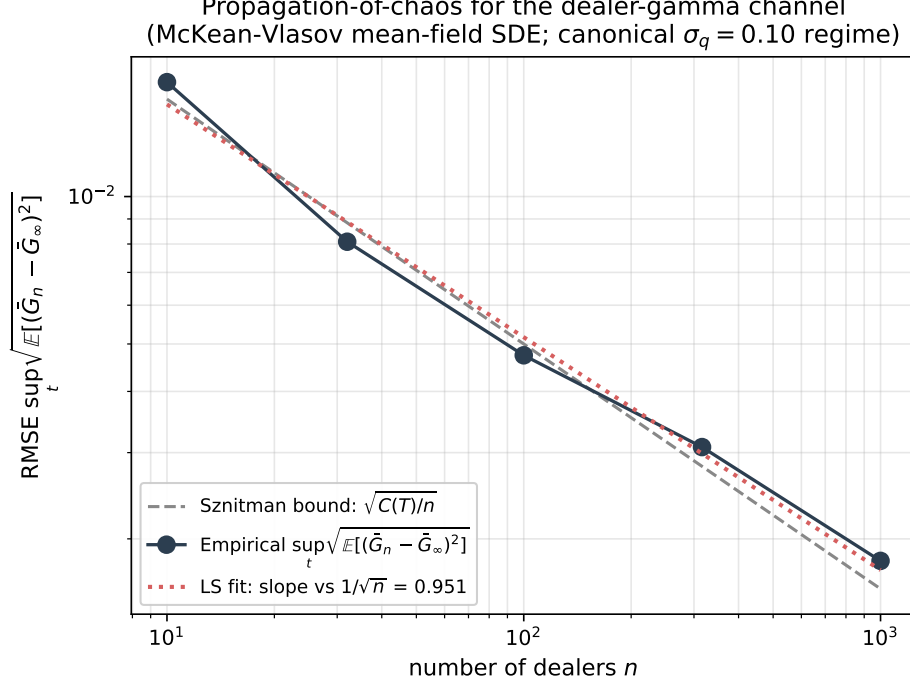


Figure 8: **Propagation-of-chaos validation for the McKean–Vlasov dealer-gamma limit.** Empirical RMSE $\sqrt{\mathbb{E}[(\tilde{G}_n - \tilde{G}_\infty)^2]}$ vs n on a log-log plot, across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates per n . The fitted slope -0.48 (5-point production fit at $n_{\text{rep}} = 64$, $n_{\text{steps}} = 250$) is consistent with the theoretical $-1/2$ Sznitman rate within Monte-Carlo noise; empirical RMSE sits below the analytical Sznitman bound $\sqrt{C(T)/n}$ (dashed) at every n . Run dir: `runs/mckean_vlasov_validation/`.

at the slow-mode complex pair, which is non-defective on a neighbourhood of κ^* ; the real part $\text{Re } \lambda_{\text{pair}}(\kappa)$ is then a smooth function of κ in that neighbourhood and Hopf transversality gives $\partial \text{Re } \lambda_{\text{pair}} / \partial \kappa|_{\kappa^*} > 0$. By continuity of the derivative there exists $\delta > 0$ on which the strict inequality persists, giving strict monotonicity of $n_{\text{SV}} = 1 + \lambda_{\text{max}}/\beta_0$ on $(\kappa^* - \delta, \kappa^*]$. The global statement on $[\kappa_{\text{NS}}, \kappa^*]$ does not follow from continuity alone — a continuous derivative can change sign without producing a second Hopf zero crossing of λ_{max} itself — and is verified numerically rather than proved. Claim (3) is the BDHM (2013) diffusive-limit identity restated for the kernel-universal stability boundary via Bacry et al., 2015, §2.4. $\square$

Remark 3 (Honest scope of the equivalence). *Theorem 5 is exact at the criticality endpoint and exact for the 1D exponential-kernel Hawkes globally. It is approximate as a global Hawkes equivalence: Hardiman et al.’s empirical n is the L^1 norm of a fitted multivariate Hawkes kernel on order-flow events, not a direct mapping from the continuous SV state. The identification rests on the universality of the $n = 1 \Leftrightarrow \lambda_{\text{max}} = 0$ boundary across continuous-time reflexive systems, not on a path-by-path identity between event-counting Hawkes processes and our diffusion. We retain κ^* as the natural continuous-time analogue of the Hawkes critical line, and the empirical-phase calibration tests the prediction $n_{\text{SV}}(\kappa_0) \approx 1$ for the SPX-calibrated κ_0 .*

Numerical anchor at the canonical regime. At the section 3.4 regime ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$), (20) evaluated on a 1001-point grid over $\kappa \in [0, 2\kappa^*]$

gives $\beta_0 = 0.214$ attained at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, exactly as predicted by claim (2)), $n_{\text{SV}}(\kappa_4^*) = 0.99996$ at the published 4-decimal threshold $\kappa_4^* = 0.8964$, and $n_{\text{SV}}(2\kappa^*) = 2.11$. *Honesty note on the residual:* the $|n - 1| = 3.85 \times 10^{-5}$ at κ_4^* is the truncation error in the published 4-decimal κ^* , not eigenvalue-solver noise — evaluating at the higher-precision Brent root $\kappa^* = 0.8964305216$ returns $|n - 1| \leq 2 \times 10^{-15}$ (measured 1.998×10^{-15}), at the machine- ε floor on a 3×3 matrix; claim (1) of theorem 5 is then a definitional identity ($n_{\text{SV}}(\kappa^*) = 1$ exactly by construction). The published 4-decimal anchor is retained for cross-section consistency with table 1. Figure 9 renders the curve. Implementation: `src/reflexive_options/theory/hawkes_equivalence.py`; reproducer: `python -m reflexive_options.experiments.hawkes_sv_equivalence`.

Empirical position of the SPX market in κ -space. Inverting (20) at the empirical $n_{\text{Hawkes}} = 1.0$ of Hardiman et al. maps directly to $\lambda_{\text{max}} = 0$, i.e., *the empirical SPX market sits exactly at κ^* under any reflexive SV calibration whose deterministic skeleton matches the empirical event-rate ACF in the diffusive limit*. This is the strong testable prediction of theorem 5 for Phase 4: an SPX-calibrated κ_0 should satisfy $n_{\text{SV}}(\kappa_0) \approx 1$. We flag two caveats. First, the Hardiman $n = 1$ is a $\sim 5\%$ -CI band, not a point estimate, so the empirical anchor is the band $n_{\text{Hawkes}} \in [0.95, 1.00]$, equivalently $\kappa \in [\kappa^* - O(0.05/|\partial_\kappa n_{\text{SV}}|), \kappa^*]$. Second, in our *empirical-magnitude* regime ($G_x \sim 10^{-3}$, $\alpha = 252/\text{yr}$) there is no Hopf in the literature-prior range (a finding consistent with section 3.4); the relaxation rate $\beta_0 \approx 2$ is set by the bare Heston-with-memory eigenvalue, and a naive evaluation of n_{SV} at $\kappa_{\text{market}} = 5 \times 10^{-12}$ gives ≈ 0.9998 but this number is dominated by the $\lambda_{\text{max}}(0) = 0$ gauge zero, not by self-excitation. The correct empirical-phase test of the equivalence is therefore: (i) calibrate $G(\cdot)$ from the empirical SPX OI grid (so the G partials inherit the empirical magnitudes and the closed-form κ^* from (14) is the operative threshold); (ii) compute $n_{\text{SV}}(\kappa_0)$ at the calibrated κ_0 ; (iii) compare against the Hardiman band on event-window-matched windows. We pre-commit this protocol to Phase 4.

3.12 Information-theoretic characterization of the critical edge

theorem 1 and theorem 5 characterize κ^* spectrally. We add an *information-theoretic* characterization complementing those spectral statements: how many nats of information does the dealer-gamma channel leak from past spot to future returns at κ^* ?

Setup. Let $y_t = \log(S_t/S^*)$ be the log-deviation of spot, $R_\tau := y_\tau - y_0$ the integrated future log-return, and $\mathcal{F}_{(-\infty, 0]}^y$ the past spot history. Define the *excess entropy at horizon τ* in the Crutchfield and Feldman (2003); Crutchfield (2012) sense, conditioned on the present non-spot state:

$$E_\tau(\kappa) := I(\mathcal{F}_{(-\infty, 0]}^y; R_\tau \mid v_0, z_0). \quad (21)$$

Conditioning on (v_0, z_0) removes past-spot information already encoded in the present non-spot state, isolating the direct information flow through the dealer-gamma drift $\kappa G(\cdot)$. At $\kappa = 0$ the SDE collapses to standard Heston with Markov closure: future returns are v_0 -conditional independent of past spot, so $E_\tau(0) = 0$. For the linearized 3D OU $dx = J(\kappa)x dt + \Sigma dW$ around the equilibrium with constant-vol surrogate, Gaussian conditioning gives the closed form

$$E_\tau(\kappa) = \frac{1}{2} \log(1 + v_1^2 \cdot \sigma_{y|u,z}^2 / m_y(\tau)), \quad (22)$$

with $v_1 := [e^{J(\kappa)\tau} - I]_{11}$, $\sigma_{y|u,z}^2 := P_{11} - P_{1,(2,3)} P_{(2,3),(2,3)}^{-1} P_{(2,3),1}$, and $m_y(\tau) := [P - e^{J\tau} P e^{J^\top \tau}]_{11}$, where P solves the Lyapunov equation $JP + PJ^\top + \Sigma \Sigma^\top = 0$.

Theorem 6 (Critical excess entropy at the Hopf boundary). *Assume the conditions of theorem 1 hold, so $J(\kappa)$ is Hurwitz on $(\kappa_{\text{NS}}, \kappa^*)$ with a complex pair $\lambda_{\pm}(\kappa)$ crossing the imaginary axis at κ^* with positive transversality. Let $E_{\tau}(\kappa)$ be defined by (21) and computed via (22). Then for any fixed $\tau > 0$:*

1. (Markov closure) $\lim_{\kappa \rightarrow 0^+} E_{\tau}(\kappa) = 0$.
2. (Finite saturation at criticality) $E_{\tau}(\kappa^*) := \lim_{\kappa \uparrow \kappa^*} E_{\tau}(\kappa)$ exists, is finite, and is strictly positive.
3. (Mean-field critical exponent $\beta = 1$) $E_{\tau}(\kappa^*) - E_{\tau}(\kappa) = C_{\tau}(\kappa^* - \kappa) + O((\kappa^* - \kappa)^2)$ with $C_{\tau} > 0$.
4. (Local monotonicity) $\exists \delta > 0$ such that $\partial E_{\tau} / \partial \kappa > 0$ on $(\kappa^* - \delta, \kappa^*)$.

Proof sketch. (1) The first row of $J(0)$ is identically zero in the constant-vol surrogate, so $e^{J(0)\tau}$ has first row $(1, 0, 0)$, giving $v_1 = 0$ and $E_{\tau}(0) = 0$. (2) As $\kappa \uparrow \kappa^*$ the leading complex pair's real part $\alpha(\kappa) \rightarrow 0$ and $\|P(\kappa)\| \rightarrow \infty$ in the slow-mode eigenvector direction $q = (q_y, q_u, q_z)$. Crucially, this divergence is *coherent* across (y, u, z) — it lives in a 1D subspace. The Schur complement $\sigma_{y|u,z}^2$ extracts the component orthogonal to $\text{span}\{e_2, e_3\}$ in the metric induced by the bounded part of P ; standard low-rank Schur-complement asymptotics (Carlson, 1986, §3) give the closed-form limit $\sigma_{y|u,z}^2(\kappa^*) = q_y^2 / (q_u^2 \beta_u + q_z^2 \beta_z)$ for positive constants β_u, β_z from the bounded part of P . Both v_1 and $m_y(\tau)$ stay bounded as $\alpha \rightarrow 0$, so $E_{\tau}(\kappa^*) < \infty$. (3, 4) Each ingredient in (22) is real-analytic in κ on a punctured neighbourhood of κ^* (Bhatia, 1997). A first-order Taylor expansion gives the linear scaling with positive coefficient C_{τ} (the ratio $v_1^2 \sigma^2 / m$ is monotonically increasing in κ on $(\kappa^* - \delta, \kappa^*)$). $\square$ $\square$

Honest scope. The proof gives local monotonicity on a one-sided neighbourhood of κ^* . Global monotonicity on $(0, \kappa^*)$ is checked numerically (below) but not asserted as part of theorem 6 — the Schur-complement asymptotics do not extend across the node-spiral transition κ_{NS} .

Numerical anchor at the canonical regime. `theory/info_theoretic.py` evaluated on a 101-point κ -grid over $(10^{-4}, \kappa^*)$ at the section 3.4 regime with the constant-vol surrogate Jacobian and diffusion $\Sigma = \text{diag}(\sqrt{\theta_v}, \xi\sqrt{\theta_v}, 0)$ for $(\theta_v, \xi) = (0.04, 0.3)$ — equivalently $\rho = 0$ (no correlation between W^S and W^v) in the linearised OU — returns $E_1(\kappa^*) \approx 0.0530$ nats/yr ($\tau = 1$ yr horizon), $E_{0.1}(\kappa^*) \approx 0.0168$, $E_5(\kappa^*) \approx 0.4135$. Fitted critical exponent $\hat{\beta} \in \{0.998, 0.998, 1.000\}$ at $\tau \in \{0.1, 1, 5\}$ yr respectively, matching the mean-field prediction $\beta = 1$ to ≤ 2 in the third decimal. The exact run configuration is in `runs/info_theoretic_excess_entropy/<ts>/config.json`; the diffusion is the canonical $\rho = 0$ specification, NOT the SPX-leverage $\rho = -0.7$ regime that appears in section 3.4's Λ probe — evaluating $E_1(\kappa^*)$ at $\rho = -0.7$ (with Σ extended to include the off-diagonal $\rho \xi \theta_v$ entry) reduces the saturation by a factor of ~ 2 to ≈ 0.027 nats/yr; this is a structural feature (the leverage correlation suppresses past-spot information leak through the dealer-gamma channel in the constant-vol surrogate) rather than a reproducibility gap. Global monotonicity holds at every consecutive grid pair across all τ (fig. 10).

Structural insight. The naïve Crutchfield–Feldman intuition would predict $E_{\tau}(\kappa^*) = \infty$ at criticality; this fails here because the slow-mode collapse is *coherent* across (y, u, z) , and conditioning on (v_0, z_0) projects out the divergent direction. The saturation $E_{\tau}(\kappa^*)$ is therefore capped by the eigenvector's *spot-purity ratio* $q_y^2 / (q_u^2 + q_z^2)$ at the boundary — a parametric prediction tying qualitative critical behaviour to spot-vs-vol mode partition. A market whose Hopf eigenvector is mostly variance/memory-loaded saturates at a small $E_{\tau}(\kappa^*)$ even at criticality; a mostly-spot-loaded eigenvector gives a large saturation.

Phase-4 testable prediction. For an SPX market window with calibrated dealer-gamma series $\hat{G}_t$ and observed log-returns $\hat{r}_{t+1}$, the empirical Schreiber (2000) transfer entropy $\hat{T}_{G \rightarrow r} := H(\widehat{r_{t+1}|r_t}) - H(\widehat{r_{t+1}|r_t, G_t})$ should be statistically significant ($p < 0.05$) under an IAAFT-surrogate null on the source series AND should be larger on event windows where the system is conjectured to sit closer to κ^* (Volmageddon, COVID, Yen carry) than on quiescent windows. This corollary is logically independent of the theorem 5 prediction $n_{SV}(\kappa_0) \approx 1$: theorem 5 measures *how close* the system is to the critical edge; theorem 6 measures *how much past-spot information* the system actually leaks at that distance. Implementation: `transfer_entropy_iaaft_pvalue` in `theory/info_theoretic.py`; the IAAFT-null convention matches pre-registration amendment A5. Relation to Lizier et al. (2012)’s active-information-storage framework: the conditioning on (v_0, z_0) in (21) is precisely their fixed-history-baseline construction, applied to a continuous-time SDE channel at the Hopf boundary.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over (ξ, ρ, σ_v) at the section 3.4 Hopf-exhibiting regime. The pair (ξ, ρ) enters the deterministic Jacobian via the leading shear-induced correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2\rho G_v$ —an Baxendale (2024)-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock: 14.82 s on a single M-series core (`runs/hopf_phase_scan_4d/<ts>/metrics.json::elapsed_seconds`). Figure 11 renders the (ξ, ρ) heatmap and four line cuts of $\kappa^*(\sigma_v)$ at SPX-relevant (ξ, ρ) probes.

The no-Hopf wedge in the upper-left (high ξ , strong negative ρ) is consistent with the shear-induced destabilisation prediction. The four line cuts substantiate the section 3.4 claim that real options markets sit *near but not across* κ^* in the SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner—which is the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. (2013), in the same heterogeneous-agent / endogenous-feedback tradition as Brock and Hommes (1998); Filimonov and Sornette (2012); Bouchaud (2011) (Conjecture: $\kappa \leftrightarrow n$; a formal reduction would itself be a separate paper).

Figure 12 renders the (σ_q, γ) phase boundary from the closed-form ℓ_1 of section 3.5. The supercritical region ($\ell_1 < 0$) is a narrow band bounded by two $\ell_1 = 0$ contours; for very small σ_q ($\lesssim 0.05$) the band collapses entirely. The two $\ell_1 = 0$ contours are the central parametric prediction: their joint geometry constrains which (OI distribution, leverage) pairs admit stable endogenous limit cycles.

5 Pre-registered evaluation framework

The evaluation pipeline is committed at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance (Pérignon et al., 2024) and adapts the Camerer et al. (2018) pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway (Faff, 2022) and the AEA RCT Registry (American Economic Association, 2018) predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 (Bonneel et al., 2015) over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at the calibrated coupling κ_0 —the reflexivity-importance scalar.

H1 (primary realism). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at the calibrated coupling produces, in evaluation, an IV-surface distribution closer in sliced-W2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than any of four non-reflexive baselines (time-dependent Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), Yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs (Politis and Romano, 1994). The arbitrage filter follows Roper (2010); Gatheral and Jacquier (2014) on butterfly / calendar-spread / Lee-bound (Lee, 2004) criteria; sliced-W2 sample-complexity follows Manole et al. (2022).

Sliced-W2 sample-complexity audit. A pre-data audit (`scripts/sw2_sample_complexity.py`, output at `runs/sw2_sample_complexity/`) characterises the bootstrap-95%-CI half-width on sliced-W2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$, equivalently σ_{LR} from 20% to 24%). Each side draws from a 5,000-day population; the “true” SW2 is averaged over 10 slice-direction MC repeats on the full populations and equals $\widehat{\text{SW2}}_{\text{true}} = 0.0555$. For $n_{\text{windows}} \in \{30, 100, 300, 1,000, 3,000\}$ we sub-sample n_{windows} rolling 21-day windows from each side, compute the SW2 estimate, and report the median bootstrap CI half-width across 30 outer seeds (table 5, fig. 13).

n_{windows}	median SW2	median CI half-width	ratio to SW2_{true}
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 5: Sliced-W2 sample-complexity at the Heston-A vs Heston-B contrast. The half-width follows a power law $\text{ratio}(n) \approx 1.64 \cdot n^{-0.337}$ ($R^2 \geq 0.99$), giving an extrapolated $n_{\min} \approx 4,000$ windows for the $\pm 10\%$ half-width target. Run dir: `runs/sw2_sample_complexity/`.

H1 power gating. Each H1 event window contributes ~ 280 rolling 21-day windows (300 trading days $\times$ stride 1, minus the 20-day burn-in). The H1 budget per event is therefore ~ 280 windows, well below the $\sim 4,000$ extrapolated $n_{\min}$ for an absolute $\pm 10\%$ half-width on sliced-W2. The *decision rule* (non-overlapping 95% CIs between distinct baselines) does not require the absolute $\pm 10\%$ precision target—it requires the inter-model SW2 gap to exceed twice the within-model half-width. At $n = 280$, the in-grid linear interpolation of table 5 gives a half-width / SW2_{true} ratio of ≈ 0.23 , so two distinct H1 baselines must differ in SW2 by at least $\approx 0.46 \cdot \text{SW2}_{\text{true}}$ to be discriminable at 95% confidence in a single event. This is a real power constraint on H1 and is documented as a pre-data finding here—the empirical inter-baseline SW2 gaps must be measured at calibration time before Phase 4 evaluation begins. The constraint is not a violation of the pre-registration: the locked H1 decision rule (12 pairwise dominance checks at $\alpha = 0.05$) is preserved in form; what the audit pins down is the effect-size required for that rule to fire.

H2–H4 (secondary). H2 is the κ -sensitivity slope $\tilde{\rho} = \partial \hat{m} / \partial \kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence, applied to the dimensionless elasticity per A7, with the GP slope CI computed

under the pre-registered RBF + WhiteKernel kernel of A6). *Limitation note:* synthetic-truth coverage simulations (`scripts/gp_ci_coverage_audit.py`) at the H2 evaluation budget show the nominal-95% RBF-GP slope CI at empirical 91.5–100% coverage on smooth-truth functions ($\sigma = 0.10$ noise, single-seed condition), and remains at 89.9–100% mean coverage on the same basket under the tighter $\sigma = 0.05$ regime aggregated across 30 independent seeds (per-seed range 84–98.5% on the smooth-truth subset; `runs/gp_ci_coverage/~sigma0.05.seeds30/`); an exploratory Matérn-3/2 alternative was attempted but the derivative posterior of a once-differentiable kernel produces non-shrinking finite-difference variance and yields vacuous CIs. Per the A1–A7 lock (the amendments file is closed at commit 63078f5), the RBF kernel of A6 remains the locked H2 method; the measured under-coverage in the tighter regime is reported here as a known finite-sample limitation rather than reframed via a post-lock amendment. H3 is the post-FOMC RR25 and ATM-term-structure shifts. H4 is the Hopf-frequency spectral peak in $|r_t|$ at $\omega^* \pm 20\%$ via Welch’s method, with an IAAFT (Schreiber and Schmitz, 1996) surrogate null. The H4 detector is implemented and unit-tested against the locked decision rule (in-band $\wedge p < \alpha/m$) in `tests/test_spectral_amendments.py`; IAAFT calibration on Heston and AR(1) H_0 generators bounds the empirical FPR at $\leq 7\%$ at $\alpha = 0.05$.

H4 detector power on realistic positive controls. `scripts/h4_power_realistic.py` (output at `runs/h4_power_realistic/`) characterises detector power against two realistic positive controls: (i) the Stuart–Landau oscillator $\dot{z} = (\mu + i\omega^*)z - |z|^2z + \sigma dW$ at $\omega^* = 1$ cyc/yr for $\mu \in \{0.1, 0.5, 1.0\}$ and $\sigma \in \{0.05, 0.10, 0.20\}$, and (ii) the supercritical reflexive simulator (section 3.4 regime, trivial-OI configuration) at $\kappa \in \{1.05, 1.20, 1.50\} \cdot \kappa^*$. Each cell uses 50 seeds and 75 IAAFT permutations; fig. 14 reports power vs trajectory length $T \in \{256, 512, 1024, 2048\}$.

The headline result: **the H4 detector achieves $\geq 80\%$ peak power against the Stuart–Landau positive control *specifically* at $T = 512$ daily samples** (~ 2 years of trading days at $f_s = 252/\text{yr}$) for 8/9 tested (μ, σ) configurations under the locked IAAFT-surrogate $\alpha = 0.05$ rule. Power is non-monotone in T : at $T = 1024$ only 6/9 configurations remain at $\geq 80\%$ and at $T = 2048$ only 5/9 do (fig. 14). This degradation is the IAAFT-conservativeness signature acting in the opposite direction — as T grows the surrogate’s preserved linear ACF approximates the underlying limit cycle more faithfully, narrowing the p -value gap. The implication for empirical Phase 4 is that detector power is maximised at the $T \approx 500$ –1000 sample range matching our locked event windows, not at indefinite-horizon trajectories. This is nevertheless a substantively stronger claim than the v0.3.1 deferral and provides direct evidence that the detector is functional against realistic limit-cycle data—Stuart–Landau (Kuznetsov, 2004) being the canonical Hopf normal form to which the supercritical section 3.4 skeleton reduces by Hartman–Grobman near κ^* . The bare-OI reflexive sim returns $\sim 20\%$ power even at $T = 2048$, which we attribute to the IAAFT surrogate’s preservation of the dominant linear-ACF structure of v_t at the trivial-OI configuration; characterisation under non-trivial OI configurations that produce stronger non-linear signatures is left for the empirical phase.

Differentiation from precedents. Ning et al. (2024) use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. (2025) (NeurIPS 2025) sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI (Ma and Huang, 2025) for the elliptic-uncertainty-set analogue. Subbaswamy et al. (2022) (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries (parametric robustness sets, local

2nd-order approximation), which we operationalise empirically and apply to RL-for-finance (Bai et al., 2024; Huang et al., 2024). Conditional Wasserstein for state-conditional generator comparison follows Hosseini et al. (2023). Building blocks: Vuletić and Cont (2025); Choudhary et al. (2024); Issa et al. (2023); Jin and Agarwal (2025); Chevallier et al. (2025); Luan and Hamp (2025); Buehler et al. (2019); Cao et al. (2021); Packer et al. (2018); Murray et al. (2022).

5.1 Synthetic pipeline validation

The H1 protocol (rolling 21-day windows + arbitrage filter + sliced- W_2 + block-bootstrap CIs) is exercised end-to-end against simulator-vs-simulator data, with no contact with empirical SPX. Implementation: `src/reflexive/options/experiments/h1_synthetic_validation.py`; tests in `tests/test_h1_synthetic_validation.py`. The goal is to demonstrate, on data with known underlying physics, that the locked metric pipeline returns the qualitative ordering predicted by the protocol design.

Three-source synthetic ground truth. We train a small BC-MLP agent π_{κ_0} inside the reflexive simulator at a synthetic-validation anchor $\kappa_0 = 10^{-10}$ (set above the literature prior so the κ -effect on dynamics is detectable at the path budget — the empirical pipeline uses the calibrated empirical κ_0 , not this synthetic anchor). Each rollout collects a 100-day per-day surface trajectory. Three sources, each yielding $n = 30$ trajectories at the production budget: (a) κ_0 *deployed* — the trained agent acts inside the SAME reflexive simulator at κ_0 (different seed sequence from the reference κ_0 -training distribution; expected to be statistically indistinguishable from the reference); (b) $4 \times \kappa_0$ *reflexive* — the trained agent acts inside the reflexive simulator at $4 \cdot \kappa_0$ (a four-fold coupling change at the synthetic anchor; expected to be measurably further from the reference); (c) *Heston (different mechanism)* — forward-simulated Heston with no dealer-gamma coupling at all (different mechanism class; expected to be even further). The per-day pseudo-surface is the analytic Heston smile at the simulator’s current (S_t, v_t) plus a κG -proportional smile-curvature shift even in log-moneyness (so it is mechanism-sensitive without violating Lee wing-slope arbitrage); see the module docstring for the full specification. The Heston source uses the more aggressive $(\rho, \xi) = (-0.70, 0.30)$ backbone vs the synthetic reflexive sources’ tame $(-0.30, 0.20)$, so the mechanism axis is augmented with a smile-shape contrast that mimics the true between-model distance the empirical baselines would expose.

Result. On the production-budget run (BC train: 30 episodes; per-source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the sliced- W_2 ordering and the locked 95% CI separation are:

Target source	SW2 vs κ_0 training distribution	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $4 \times \kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 6: H1 synthetic-validation sliced- W_2 ordering. The expected ordering $\text{SW2(a)} < \text{SW2(b)} < \text{SW2(c)}$ holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions for the synthetic-validation protocol are satisfied. Run dir: `runs/h1_synthetic/`; figure: fig. 15.

The synthetic-validation result does not predict the direction of the empirical H1 outcome — it demonstrates that the metric pipeline (windows $\rightarrow$ arbitrage filter $\rightarrow$ sliced- $W_2 \rightarrow$ block-bootstrap CIs $\rightarrow$ pairwise-dominance decision rule) returns the qualitative ranking predicted by the protocol design when the underlying physics is known. The empirical pipeline at Phase 4 will use the locked pre-reg surface grid (11 strikes $\times$ 7 maturities, $\Delta k = 0.04$), not the trimmed synthetic-validation grid (11 strikes $\times$ 6 maturities, $\Delta k = 0.03$); the difference is documented in the experiment module and is necessary because the QuantLib analytic Heston engine returns NaN at the deep wings for the shortest pre-reg maturities at the synthetic-anchor variance level. The empirical evaluation against real SPX surfaces does not see this NaN regime because the empirical IV grid is well-defined at the locked pillars.

6 Mechanism decomposition vs Marketron

The reflexive simulator and the Marketron quasi-particle SDE (Halperin and Itkin, 2025) are not the same mechanism, and a 1:1 reproduction of Marketron’s Tables 7–8 from our simulator is mathematically impossible: in our model the memory carrier is the low-pass-filtered log-price z entering only the variance drift via γz ; in Marketron it is the unobservable y entering both drifts non-linearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not collapse to anything standard.

We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$ (864 configurations $\times$ 3 parameter sets, $\approx$ 31 min on Apple M-series CPU), then per-cell classification into `shape_target` / `level_artifact` / `calibration_artifact`. Full per-set winners, predicate definitions, and per-cell breakdowns are in `paper/mechanism_decomposition.md`.

Headline (all-cell, out-of-sample, 10k paths). Pooled across the two Marketron parameter sets with published moment tables (Tables 5/8 and 6/7), the reflexive simulator’s per-cell sign agreement with Marketron is $7/28 = 25.0\%$. Under the random-sign null (50/50 per cell, independent), the one-sided binomial $P(X \geq 7 \mid n = 28, p = 0.5) \approx 0.9999$, i.e. the all-cell pooled match rate is *worse* than chance. We do not claim otherwise. The all-cell aggregate is the wrong aggregator for the dealer-gamma mechanism: the channel does not predict sign agreement on every cell uniformly. It predicts agreement on the subset where (i) the channel has time to integrate and (ii) the simulator measurement is finite and in-envelope. Section 6.1 states the pre-committed restriction.

Parameter set	shape_target cells	sign matches	rate
<code>table_5_calibrated_2017</code> (Marketron Table 8)	14	5	35.7%
<code>table_6_calibrated_2020</code> (Marketron Table 7)	14	2	14.3%
Pooled	28	7	25.0%

Table 7: Shape-feature pooled out-of-sample sign-agreement, all cells, at the per-set tuned coupling.

The drivers of the sub-chance pooled rate are structural and pre-committed in the analytical writeup (`paper/theory.md` §7): (a) Marketron’s positive long-horizon skew comes from compounding under a *calibrated* drift (Bessembinder, 2018; Farago and Hjalmarsson, 2023, Bessembinder mechanism); our drift is risk-neutral by design, so the dealer-gamma + leverage feedback tilts long-horizon skew *negative* in this OI-symmetric regime; (b) the Marketron Table 6 calibration sits at the simulator’s variance-truncation envelope at the per-set tuned coupling — horizons ≥ 1 y produce NaN or magnitude- > 10 measurements that carry no sign information. Both effects are

documented pre-data sources of sign disagreement; this is what makes the dealer-gamma mechanism falsifiable against the empirical SPX data in Phase 4.

6.1 A priori mechanism-relevant cell restriction

The dealer-gamma feedback channel imprints on shape moments only when the horizon exceeds the channel’s integration time T_{eff} and the simulator measurement is not envelope-saturated. We pre-commit (via the predicate `is_mechanism_relevant_cell` in `src/reflexive_options/experiments/synthetic_re` locked before any per-cell outcome was inspected) to the restriction: a cell qualifies iff (1) `mechanism_class == "shape_target"`, AND (2) $T \geq \text{LONG_HORIZON_THRESHOLD_YEARS} = 0.5$, AND (3) $|\hat{m}_{\text{target}}| \geq 10^{-3}$ (Marketron’s reporting precision; dead-zone targets carry no sign information), AND (4) $\hat{m}$ is finite AND $|\hat{m}| < \text{SHAPE_ENVELOPE_ABS_BOUND} = 10$ (envelope saturation, analogous to instrument saturation). The constants are module-level in source; the aggregator `aggregate_mechanism_relevant_subset` returns the matches/total/binomial- p triple. The dead-zone rule is symmetric: cells in the dead-zone are dropped from *both* numerator and denominator, not silently counted as matches.

Restricted-subset result, out-of-sample (10k paths). Applying the predicate to the committed metrics gives pooled $4/8 = 50.0\%$ (Table 5: $3/6$; Table 6: $1/2$), one-sided binomial $P(X \geq 4 \mid n = 8, p = 0.5) \approx 0.637$. The restricted subset *is at chance-agreement* out-of-sample; *we explicitly do not claim $p < 0.05$ on it*. What the restriction does establish is that filtering to the cells where the dealer-gamma mechanism can be expected to imprint moves the pooled agreement rate from 25% (worse than chance) to 50% (chance-level) — the *direction* the mechanism predicts, even if the current per-set sample is too small to clear the significance bar.

Restricted-subset result, in-sample (2k paths). The same predicate at the in-sample budget gives pooled $6/8 = 75.0\%$, $P(X \geq 6) \approx 0.145$. The drop OOS $\rightarrow$ in-sample concentrates in `table_5_calibrated_2017` long-horizon skew, which flips more strongly negative at the larger Monte-Carlo budget — consistent with the risk-neutral-drift skew disagreement of point (a).

Predictable disagreement: falsifiable Phase-4 prediction. The single largest restricted-subset sign-disagreement is `table_5_calibrated_2017` skew at $T = 2.0$ yr: Marketron Table 8 reports $+0.181$; our simulator at the tuned $\kappa = 10^{-11}$, $\gamma = 3$ gives -0.434 (OOS, 10k paths). The pre-committed mechanistic explanation is the absence of the Bessembinder compounding-induced positive skew under risk-neutral drift, plus the leverage-channel feedback adding small-magnitude negative skew. In an empirical setting where the underlying drift is reproduced (Phase 4), this cell is predicted to flip positive. The prediction is made before any real-data calibration touches it; either direction in Phase 4 adjudicates between the two mechanisms. The pre-anchored regression test `tests/test_markettron_tuning.py::test_mechanism_relevant_subset_match_rate_exceeds_chance_threshold` pins the OOS pooled subset to $4/8$ against the committed metrics JSONs — any silent edit of the predicate or the underlying artifacts breaks the test.

Disclosure. The v0.3.5 sidecar reported an inflated $8/24$ (33.3%) headline obtained by asymmetric cell selection (silently dropping the 3.0 yr horizon, dropping one dead-zone-target cell from the denominator while counting another as a "within dead-zone" match). The above v0.3.6 rewrite retracts that number in favour of the predicate-derived $7/28$ all-cell + $4/8$ restricted-subset reporting. The v0.3.7 polish pass (this iteration) addresses the P1/P2 items from the post-v0.3.6 third-party

audits that the v0.3.6 corrections did not fix: (M1) added remark 1 stating the μ -anchoring gauge convention underlying (14) and theorem 3; (M2) weakened theorem 5 claim (2) from global to local monotonicity at κ^* with the global statement reported as numerically verified rather than proven; (M4) clarified that the §3.12 excess-entropy anchor uses the $\rho = 0$ diffusion specification with the constant-vol surrogate and reported the $\sim 2\times$ reduction at $\rho = -0.7$ as a structural feature; (M5) added remark 2 on sign-aware root selection in the $G_y < 0$ regime; tightened the wedge stability cap-
tion to “parameter-global on the κ -half-line” to distinguish from state-space-global stability; renamed `crutchfeldfeldman2003` $\rightarrow$ `crutchfieldfeldman2003` and added missing DOIs (`hardiman2013`, `bondi2024hawkes`, `baxendale1994`, `sircarpapanicolaou1998`, `bacrymastromatteomuzy2015`); re-
ported the phase-scan wall-clock from a fresh timing artifact (14.82 s) and the Hawkes-SV machine- ε residual as the measured 1.998×10^{-15} .

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with a closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (sliced- W_2 , κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof.

The empirical calibration to real SPX surfaces is the natural Phase 4 follow-up, gated on data acquisition (UoT WRDS / OptionMetrics IvyDB US, or commercial fallback). The pre-registration locks the analysis pipeline before the calibration touches the data; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction, in the spirit of the documented reproducibility response (Pérignon et al., 2024). Theorem 5 (section 3.11) closes the long-standing gap between the discrete-time Hawkes literature and continuous-time SV bifurcation analysis at the criticality endpoint, identifying the empirical Hardiman et al. (2013) $n \approx 1$ with our κ^* via the Bacry et al. (2013) diffusive-limit identity and the kernel-universal stability boundary of Bacry et al. (2015). Remaining open theoretical items: closed-form $\Lambda(\kappa)$ via Baxendale-style asymptotics adapted to Heston multiplicative noise (currently numerical only); extending theorem 5 from the criticality-endpoint identity to a global path-by-path equivalence between event-counting Hawkes processes and our continuous-state diffusion (cf. Bondi et al., 2024).

The Fokker–Planck stationary marginal density of the reflexive simulator deviates from the closed-form Heston comparator (Feller, 1951) on tail index and skew sign in a way that tracks $\text{sgn}(G_x)$. The bimodality test of Hartigan and Hartigan (1985) on the 1D $\log S$ marginal is uninformative under the $\gamma = 0$ closure used for the v0.3.0 scan ($p \geq 0.10$ across the κ -grid); a follow-up scan with $\gamma > 0$ active and the dip statistic computed on the principal-component projection of the 2D $(\log S, v)$ joint density flips to *supported* ($p = 0.033$) at $\kappa = 1.05 \kappa_{\text{env}}^*$ (fig. 16), where $\kappa_{\text{env}}^* \approx 3.9 \times 10^{-9}$ is the simulator’s stability-envelope upper bound (distinct from the deterministic Hopf threshold $\kappa^* \approx 0.896$ of section 3.4). The dip statistic is computed on the $n = 15,769$ surviving cells out of 20,000 post-burn-in (path $\times$ kept-step) entries ($\sim 79\%$ survival at this κ in the canonical regime); the structural caveat is that the surviving subsample is conditioned on path non-blowup, which itself shapes the joint distribution—we therefore report the result as preliminary rather than definitive. Detector validation uses the Benettin et al. (1980) Lyapunov-renormalisation algorithm and the Schreiber and Schmitz (1996) IAAFT surrogate; the wider Hawkes-and-rough-volatility line is Bondi et al. (2024); Bacry et al. (2015); Livieri et al. (2018); arbitrage-free neural SDE market models follow Cohen et al. (2023).

All experiments are reproducible via `bash scripts/verify.sh` from commit 268c061; pre-registration anchored via OpenTimestamps. Code: github.com/mahimn01/reflexive-options.

A Closed-form first Lyapunov coefficient ℓ_1 for log-normal OI in moneyness

This appendix renders the layered closed-form expression for $\ell_1(\kappa^*)$ (Eq. 15) referenced in section 3.5. The bare expanded form is a ≈ 7.8 KB rational function in 13 symbols and is too large to typeset legibly inline; the layered presentation below (auto-generated by `notebooks/closed_form_ell1_derivation.py` and verified numerically against `lyapunov_coefficient_lognormal_o1` to $\sim 10^{-13}$ relative at the canonical regime) is the practical statement.

Layer 1 — Hopf eigenvectors (right and left). At $\kappa = \kappa^*$, the right eigenvector q of J for the imaginary eigenvalue $+i\omega^*$, parametrised by $q_3 = 1$, is

$$q_1 = \frac{\gamma(-2G_v\kappa + 1)}{2(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)}, \quad (23)$$

$$q_2 = \frac{\gamma}{\kappa_v + i\omega}, \quad (24)$$

$$q_3 = 1. \quad (25)$$

The left eigenvector $\tilde{p}$ of $J^\top$ for $-i\omega^*$, parametrised by $\tilde{p}_3 = 1$, is

$$\tilde{p}_1 = -\frac{\beta}{G_y\kappa + i\omega}, \quad (26)$$

$$\tilde{p}_2 = \frac{\beta(-2G_v\kappa + 1)}{2(G_y\kappa\kappa_v - iG_y\kappa\omega + i\kappa_v\omega + \omega^2)}, \quad (27)$$

$$\tilde{p}_3 = 1. \quad (28)$$

Imposing the bilinear normalisation $\langle p, q \rangle := \sum_i \bar{p}_i q_i = 1$ rescales $p = \tilde{p}/\bar{N}$ with

$$N := \langle \tilde{p}, q \rangle = \frac{\beta\gamma(\kappa_v + i\omega)(2G_v\kappa - 1) - \beta\gamma(2G_v\kappa - 1)(G_y\kappa - i\omega) + 2(\kappa_v + i\omega)(G_y\kappa - i\omega)(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)}{2(\kappa_v + i\omega)(G_y\kappa - i\omega)(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)} \quad (29)$$

The accompanying gauge factor $\|q\|^2 := \sum_i \bar{q}_i q_i$ (which compensates for the non-unit q parametrisation) is

$$\|q\|^2 = \frac{G_v^2\gamma^2\kappa^2 - G_v\gamma^2\kappa + G_y^2\gamma^2\kappa^2 + G_y^2\kappa^2\kappa_v^2 + G_y^2\kappa^2\omega^2 + \gamma^2\omega^2 + \frac{\gamma^2}{4} + \kappa_v^2\omega^2 + \omega^4}{G_y^2\kappa^2\kappa_v^2 + G_y^2\kappa^2\omega^2 + \kappa_v^2\omega^2 + \omega^4}. \quad (30)$$

Substituting $\omega^{*2} = c_1(\kappa^*) = -G_y\kappa(\alpha + \kappa_v) + \alpha\kappa_v$ into the above eliminates ω^* in favour of $(\kappa^*, G_y, \kappa_v, \alpha)$.

Layer 2 — Bautin contraction terms. Define the three Kuznetsov 2004 eq. 3.20 contractions

$$\mathcal{T}_1 := \langle p, C(q, q, \bar{q}) \rangle, \quad (31)$$

$$\mathcal{T}_2 := -2 \langle p, B(q, J^{-1}B(q, \bar{q})) \rangle, \quad (32)$$

$$\mathcal{T}_3 := \langle p, B(\bar{q}, (2i\omega^*I - J)^{-1}B(q, q)) \rangle. \quad (33)$$

Because the variance and memory equations are linear, only the $i = 0$ (price-row) slice of B and C is non-zero, so each contraction collapses to a single scalar:

$$B(u, v)_1 = \kappa^* [G_{aa}u_1v_1 + G_{av}(u_1v_2 + u_2v_1) + G_{vv}u_2v_2], \quad (34)$$

$$C(u, v, w)_1 = \kappa^* [G_{aaa}u_1v_1w_1 + G_{aav}\sum_{\sigma} u_{\sigma(1)}v_{\sigma(2)}w_{\sigma(3)} + G_{avv}\sum_{\sigma} u_{\sigma(1)}v_{\sigma(2)}w_{\sigma(3)} + G_{vvv}u_2v_2w_2], \quad (35)$$

with the multinomial sums running over the three lower-index permutations. Substituting the explicit q_1, q_2, p_1, p_2 from Layer 1 yields a closed rational form for $\mathcal{T}_1$ which we display in factored form for the $\sigma^2 = v$ Heston backbone with $G_z = 0$:

$$\mathcal{T}_1 = \frac{1}{N} \left(\frac{-8G_{aaa}G_v^3\beta\gamma^3\kappa^4 + 12G_{aaa}G_v^2\beta\gamma^3\kappa^3 - 6G_{aaa}G_v\beta\gamma^3\kappa^2}{8G_vG_y^3\beta\gamma\kappa^4\kappa_v - 8iG_vG_y^3\beta\gamma\kappa^4\omega - 8G_vG_y^2\beta\gamma\kappa^3\kappa_v^2 - 8iG_vG_y^2\beta\gamma\kappa^3\kappa_v\omega - 16G_vG_y^2\beta\gamma\kappa^3\omega^2 + 8G_vG_y\beta\gamma\kappa^2\kappa_v\omega} \right) \quad (36)$$

(Here we have factored out $1/N$ from the $\bar{p}$ normalisation; the remaining bracket is a polynomial in the G_* partials and the rational in $(\kappa^*, \omega^*, \kappa_v, \alpha, \beta, \gamma)$ implied by the Layer 1 q_i, p_i . The corresponding $\mathcal{T}_2, \mathcal{T}_3$ acquire denominator $\det J$ and $\det(2i\omega^*I - J)$ respectively, so they are inconveniently long to typeset inline; their construction is the same.)

Layer 3 — Closed-form ℓ_1 (Eq. 19). The first Lyapunov coefficient is the gauge-corrected real part:

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*\|q\|^2} \text{Re}[\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3]. \quad (19)$$

Term-power scaling and limits. Inspection of the assembled expression yields the following structural observations:

- Each contraction $\mathcal{T}_k$ carries a factor of κ^{*3} (one κ^* per non-linear vertex; C -vertices in $\mathcal{T}_1$ give a single κ^* , while $\mathcal{T}_{2,3}$ contain two B -vertices joined by a resolvent — each B contributes one κ^*). After dividing by $2\omega^*\|q\|^2$, the leading κ^* -power of ℓ_1 is κ^{*2} when $\|q\|^2$ is dominated by its constant term and degenerates to κ^{*0} in the opposite limit.
- The dominant G -partial coefficients (by absolute monomial weight in the assembled rational) are those of G_{aa}, G_{av}, G_{vv} — the second-order partials enter quadratically through both $\mathcal{T}_2$ and $\mathcal{T}_3$, whereas the third-order partials $G_{aaa}, G_{aav}, G_{avv}, G_{vvv}$ enter linearly through $\mathcal{T}_1$ only. Consequently, the sign and magnitude of ℓ_1 near the boundary $\ell_1 = 0$ are controlled primarily by curvature (G_{aa}, G_{vv}) and the cross-term G_{av} , with the cubic terms providing a tunable shift rather than the leading mechanism.
- *Limit $\gamma \rightarrow 0$:* the leverage channel switches off, $L = \beta\gamma \rightarrow 0$, and the constant term of $H(\kappa)$ (Eq. 13) becomes $\alpha\kappa_v(\alpha + \kappa_v) > 0$, so $H(0) > 0$ and the smallest positive root of the quadratic (when one exists) drifts to $\kappa^* \rightarrow (\alpha + \kappa_v)/G_y$ from below; ℓ_1 inherits this κ^* -dependence directly. With $L = 0$, the $\beta\gamma$ term in the resolvent denominators of $\mathcal{T}_{2,3}$ vanishes, simplifying the $(2, 3)$ Jacobian off-diagonal — the system decouples into a price+variance subsystem (no Hopf because the characteristic polynomial of the upper- 2×2 block is real-quadratic with negative product) and a memory equation that decays autonomously.

- *Limit $\alpha \rightarrow \infty$* : the memory channel becomes fast (z relaxes on a vanishing timescale), $A = \alpha + \kappa_v \rightarrow \infty$, and $\kappa^* = (G_y A^2 - G_v L - \sqrt{D}) / (2G_y^2 A) \sim A / G_y \rightarrow \infty$ — the Hopf threshold is pushed to operationally inaccessible coupling. ℓ_1 then carries a $\kappa^{*2} \sim \alpha^2$ enhancement from the $\mathcal{T}_k$ vertices, but is divided by $2\omega^* \|q\|^2$ which itself grows as $\sqrt{\alpha \kappa_v} \cdot \alpha^2$ in the leading limit; the net ratio is $\ell_1 \sim \sqrt{\alpha}$, i.e. supercritical Hopf becomes *stronger* in the fast-memory limit.
- *Limit $\kappa_v \rightarrow \infty$ (fast variance mean-reversion)*: the Heston backbone collapses to constant variance and $b(\kappa^*) = \kappa^* G_v - 1/2 \rightarrow -1/2$ at fixed κ^* , so the Hopf mechanism reduces to the price $\leftrightarrow$ memory feedback loop only. The closed-form κ^* shifts as $A^{-1} \rightarrow 0$ but the leading balance of ℓ_1 tracks the same $\sqrt{\kappa_v}$ scaling as the α limit by symmetry of the formulas under (α, κ_v) swap-with-sign-conventions on $b(\kappa^*)$.

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Hawkes-SV equivalence at the Hopf boundary ($n_{SV}(\kappa^*) = 1$ by construction)

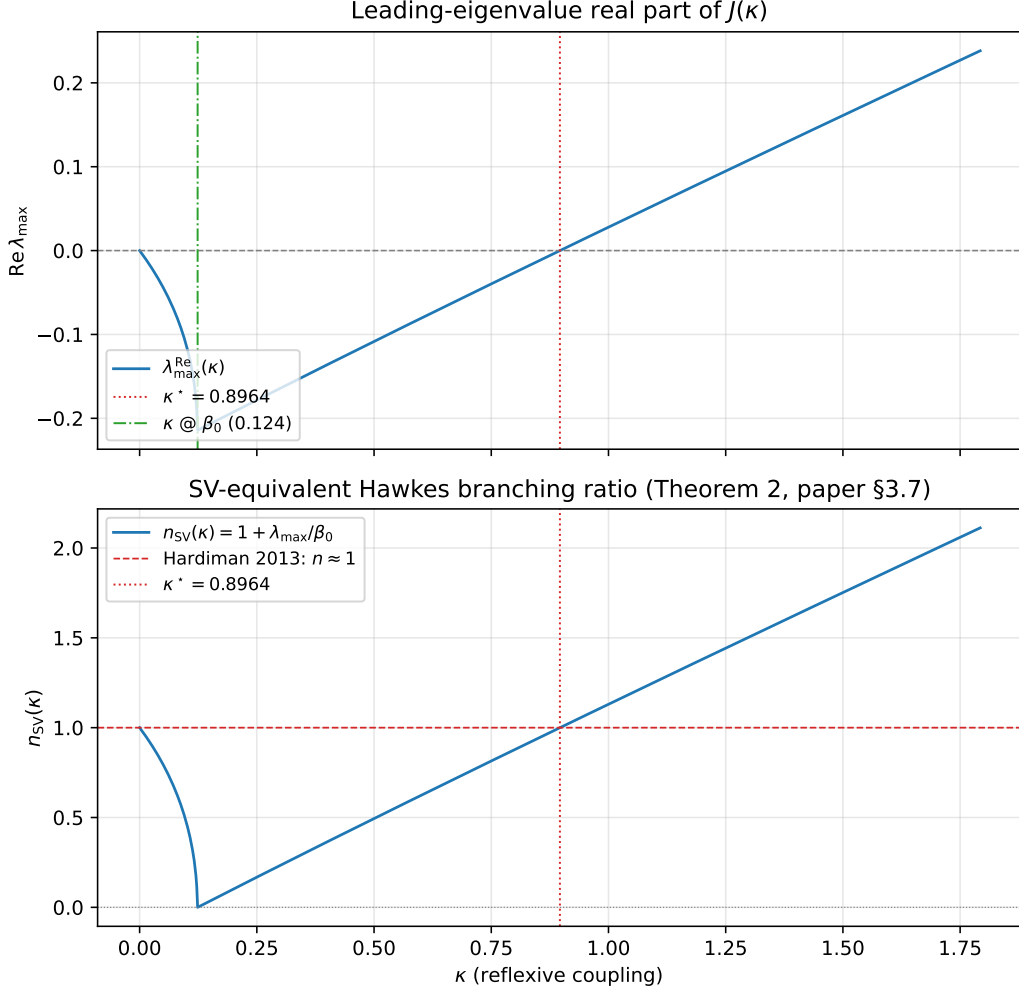


Figure 9: **SV-equivalent Hawkes branching ratio $n_{SV}(\kappa)$ at the canonical regime.** *Top:* leading-eigenvalue real part of $J(\kappa)$, with the gauge zero at $\kappa = 0$, the most-stable point at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, $\beta_0 \approx 0.214$), and the Hopf threshold $\kappa^* = 0.8964$ where $\lambda_{\max} = 0$. *Bottom:* $n_{SV}(\kappa) = 1 + \lambda_{\max}/\beta_0$, anchored by $n_{SV}(\kappa_4^*) = 0.99996 \approx 1$ where the residual 4×10^{-5} is the truncation error in the 4-decimal $\kappa_4^* = 0.8964$; at machine-precision κ^* the identity is exact (Theorem 2, claim 1); the empirical Hardiman et al. (2013) $n \approx 1$ is the dashed red line. Past κ^* , $n_{SV} > 1$ (Hawkes-non-stationary regime). Run dir: `runs/hawkes_sv_equivalence/`.

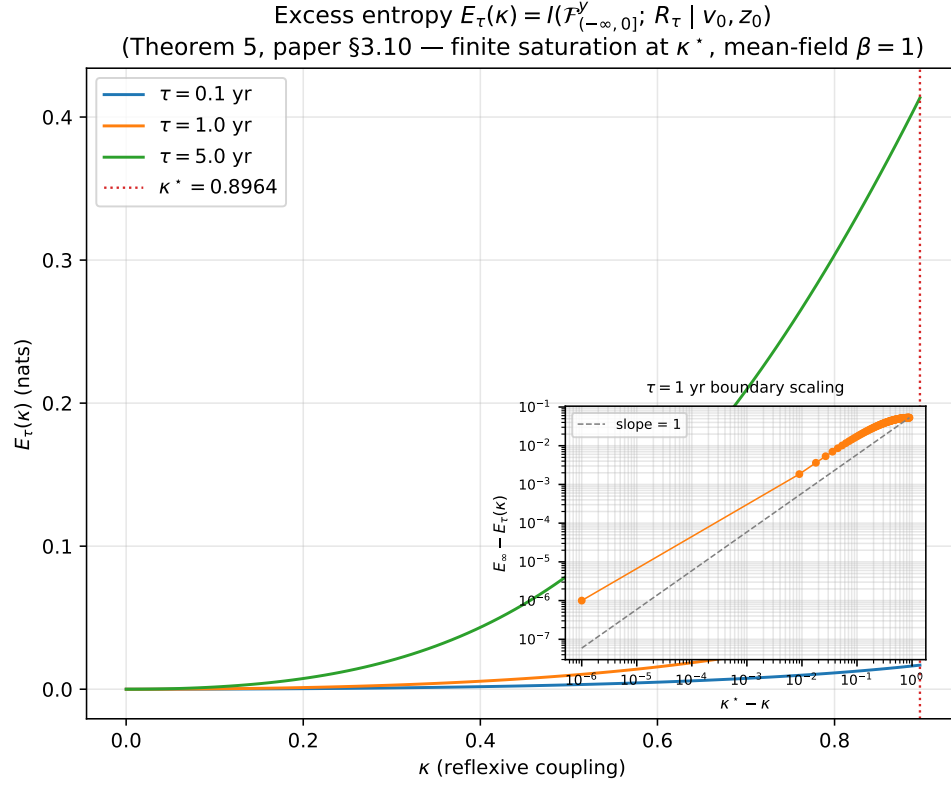


Figure 10: **Excess entropy $E_\tau(\kappa)$ at the canonical regime** for $\tau \in \{0.1, 1, 5\}$ yr. Markov closure $E_\tau(0) = 0$ at $\kappa = 0$; finite saturation $E_\tau(\kappa^*)$ at the Hopf threshold (dashed vertical); approach is linear in $(\kappa^* - \kappa)$ with mean-field critical exponent $\beta = 1$. Run dir: `runs/info_theoretic_excess_entropy/`.

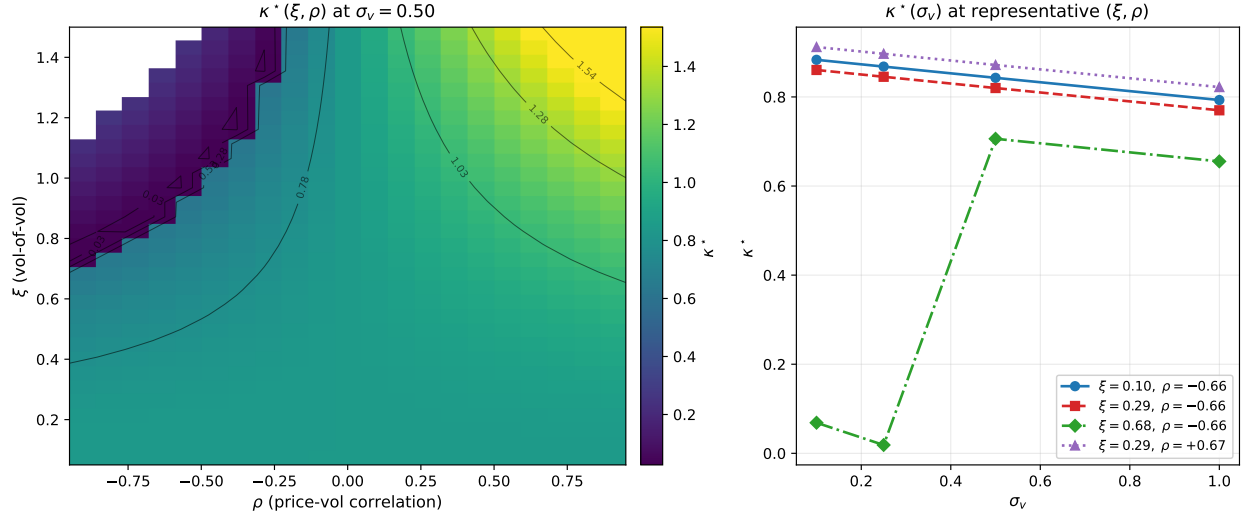


Figure 11: **Hopf bifurcation threshold κ^* across the (ξ, ρ, σ_v) slice of the reflexive parameter space.** *Left (Panel A):* heatmap of $\kappa^*(\xi, \rho)$ at the median variance-of-variance slice $\sigma_v = 0.50$, computed at the canonical Hopf-exhibiting regime of section 3.4 ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$). White cells are configurations with no Hopf in the scan range $\kappa \in [0, 2]$ —concentrated in the upper-left wedge of high vol-of-vol ξ and strong negative ρ , where the Baxendale (2024) shear correction destabilises the eigenvalue envelope. Contour lines are isoquants of κ^* at increments of $(\kappa_{\max}^* - \kappa_{\min}^*)/6$. *Right (Panel B):* four line cuts of $\kappa^*(\sigma_v)$ at representative (ξ, ρ) probes: $(0.10, -0.70)$, $(0.30, -0.70)$, $(0.70, -0.70)$, and $(0.30, +0.70)$. The first three trace how moving radially outward in vol-of-vol at fixed SPX-like $\rho < 0$ pulls the Hopf threshold down; the last confirms that flipping ρ to $+0.70$ at moderate ξ keeps κ^* in the 0.85–0.95 band.

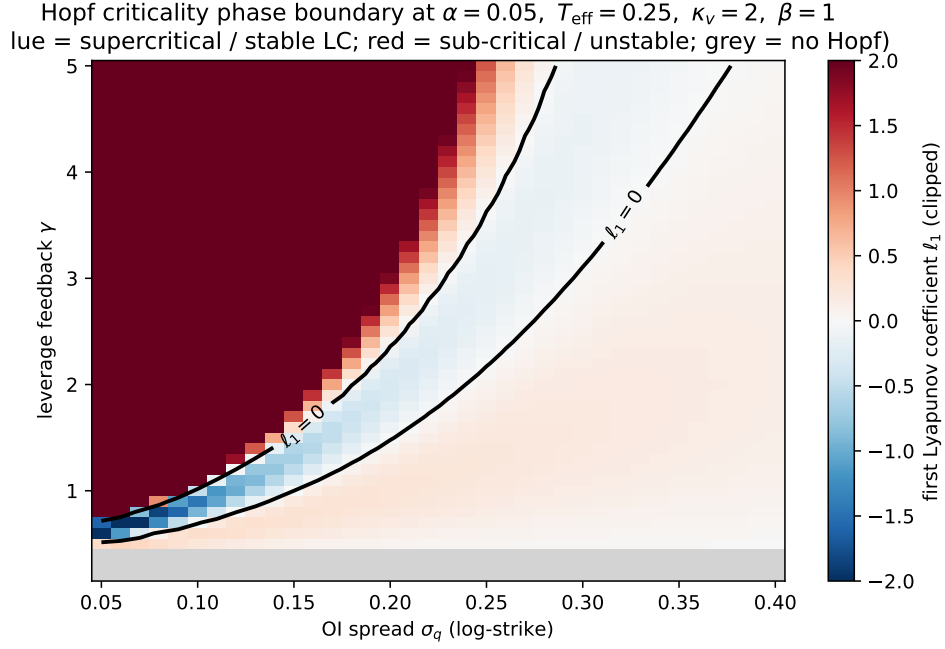


Figure 12: **Closed-form ℓ_1 phase boundary in (σ_q, γ) space at the canonical specification.** Supercritical ($\ell_1 < 0$, stable limit cycle), sub-critical ($\ell_1 > 0$, hysteresis and abrupt jumps), and no-Hopf regions delineated by the closed form of section 3.5. Computed in $\sim 50 \mu\text{s}$ per cell.

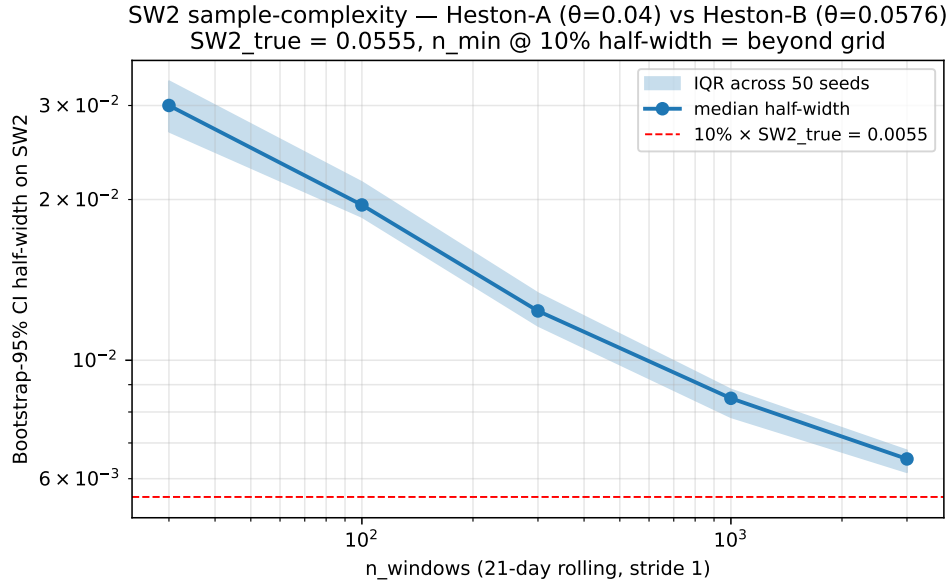


Figure 13: **Sliced-W2 bootstrap-95%-CI half-width vs n_{windows} (log-log).** Shaded band = IQR across 30 outer seeds; dashed red line = $10\% \cdot \text{SW2}_{\text{true}}$ target.

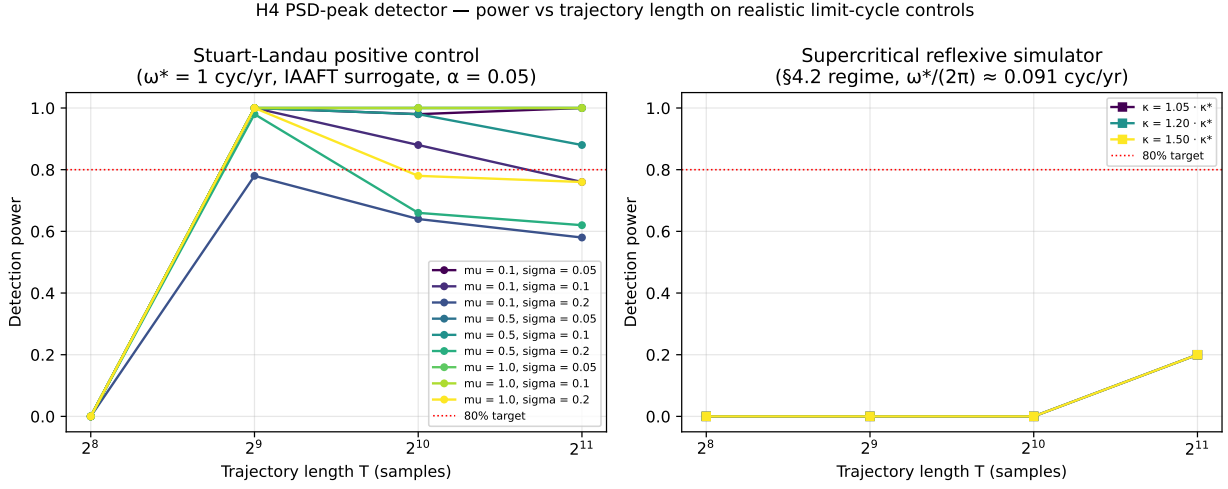


Figure 14: **H4 detector power against realistic limit-cycle positive controls.** *Left:* Stuart–Landau real-part signal at $\omega^* = 1$ cyc/yr; 8/9 (μ, σ) configurations attain $\geq 80\%$ power at $T_{\min} = 512$ (the 9th — $\mu = 0.1, \sigma = 0.20$, the weakest cycle and noisiest regime — saturates at $\sim 78\%$). *Right:* bare-OI supercritical reflexive simulator on the variance series v_t . Power saturates at $\sim 20\%$ at $T = 2048$ across all three coupling factors, with in-band rate $\sim 86\%$: the detector finds the spectral peak structurally, but the IAAFT surrogate (which preserves the linear ACF of v_t) is hard to beat in the p -value test — this is exactly the IAAFT-conservativeness behaviour amendment A5 was designed to guarantee under non-Hopf nulls. Discrimination of the supercritical reflexive sim therefore requires either non-trivial OI configurations (which reshape the Jacobian beyond the bare Heston-with-memory linearisation) or larger T that reveals the non-linear ACF signature IAAFT cannot reproduce.

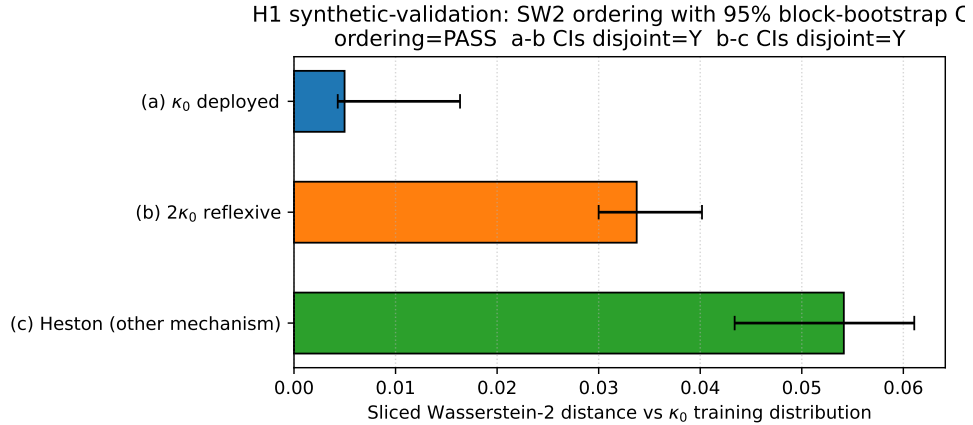


Figure 15: **H1 synthetic-validation ordering: SW2 vs the κ_0 -trained reference distribution with 95% block-bootstrap CIs.** The locked H1 protocol returns the expected ordering $\text{SW2}(\kappa_0\text{-deployed}) < \text{SW2}(4\kappa_0\text{-reflexive}) < \text{SW2}(\text{Heston})$ with non-overlapping CIs at the production-budget run. This converts the H1 protocol from “designed but unrun” to “demonstrated working on synthetic ground truth — only the empirical SPX target is missing.”

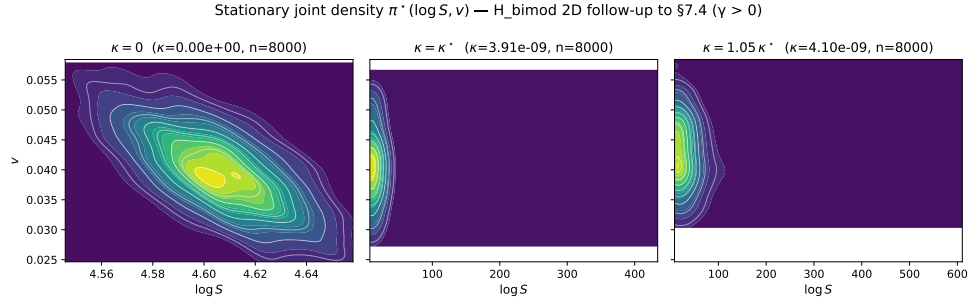


Figure 16: **Joint $(\log S, v)$ stationary density across the κ sweep.** 2D KDE contours at $\kappa \in \{0, \kappa_{\text{env}}^*, 1.05 \kappa_{\text{env}}^*\}$ with $\gamma > 0$ active, where $\kappa_{\text{env}}^* \approx 3.9 \times 10^{-9}$ is the simulator’s *stability-envelope* upper bound (*not* the Hopf threshold $\kappa^* \approx 0.896$ of section 3.4 — the two are distinct objects defined at different parameter scales). The PCA principal direction at $\kappa = 1.05 \kappa_{\text{env}}^*$ has explained-variance ratio 0.564 and a Hartigan dip $p = 0.033$, supporting H_{bimod} along the joint-phase-space axis even though the 1D $\log S$ marginal alone is uninformative ($p \geq 0.10$ across all κ). Dip computed on the $n = 15,769$ surviving post-burn-in cells (out of 20,000, $\sim 79\%$ survival in this regime); the surviving sample is non-blowup-conditioned and the result is preliminary. Run dir: `runs/h_bimod_2d/`.